

Variance of the Random k -SAT Hitting Time

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September 27, 2026

Abstract

Let H_n be the index of the first unsatisfiable prefix in random k -SAT on n variables, with independent uniformly signed clauses using k distinct variables and sampled with replacement. For every fixed $k \geq 4$, we prove $\text{Var}(H_n) = \Theta_k(n)$. For $k = 3$, the variance is bounded below by a positive multiple of n and above by a constant multiple of $n \log n$; the companion paper on random 3-SAT sharpens this to $\Theta(n)$. The same upper bounds proved here hold after clipping at any fixed positive multiple of n .

1 Fluctuations of the satisfiability transition

Add independent random clauses to a Boolean formula until the formula becomes unsatisfiable. How much does the number of clauses at this first failure vary from one sample to another? For every fixed clause size $k \geq 4$, we prove that its variance has order n , where n is the number of variables. For three-literal clauses the variance lies between positive constant multiples of n and $n \log n$.

A *proper k -clause* is a disjunction of literals on k distinct variables. Fix $k \geq 3$ and $n \geq k$, and let $C_1, C_2, \dots$ be independent uniform choices from the $2^k \binom{n}{k}$ proper signed clauses on n variables. Signs are independent and fair; whole clauses may repeat. Put

$$F_m = C_1 \wedge \dots \wedge C_m, \quad F_0 = \text{true}, \quad H_n = \min\{m \geq 1 : F_m \text{ is unsatisfiable}\}.$$

For a fixed $B > 0$, the capped first failure time is $T_{n,B} = \min\{H_n, \lfloor Bn \rfloor\}$. Each assignment survives a clause with probability $1 - 2^{-k}$, so

$$\Pr(H_n > m) \leq 2^n (1 - 2^{-k})^m. \quad (1.1)$$

Thus H_n is almost surely finite and has finite moments.

Theorem 1.1. *Fix an integer $k \geq 3$, and write*

$$\ell_k(n) = \begin{cases} \log(en), & k = 3, \\ 1, & k \geq 4, \end{cases} \quad U_k = \frac{\log 2}{-\log(1 - 2^{-k})}.$$

There are constants $C_k, c_k > 0$ such that

$$\text{Var}(H_n) \leq C_k n \ell_k(n) \quad (n \geq k), \quad \text{Var}(H_n) \geq c_k n \quad \text{for all sufficiently large } n.$$

For every fixed $B > 0$ there is $C_{k,B} < \infty$ such that

$$\text{Var}(T_{n,B}) \leq C_{k,B} n \ell_k(n) \quad (n \geq k).$$

If $B > U_k$, then also $\text{Var}(T_{n,B}) \geq c_k n$ for all sufficiently large n , with the size cutoff allowed to depend on k and B .

All constants concern fixed k . The lower condition on the cap is substantive: an early cap can stop the process before its random transition. In the proof, it is enough to have integer caps $L_n = O(n)$ with $\Pr(H_n > L_n - 1) \leq 1/4$ eventually; $B > U_k$ is an explicit sufficient condition. This argument leaves a logarithmic gap for three-literal clauses; the companion paper [17] closes it and proves $\text{Var}(H_n) = \Theta(n)$ for the uncapped proper 3-clause process.

The identity $\Pr(F_m \text{ is satisfiable}) = \Pr(H_n > m)$ turns these variance bounds into bounds on the transition window. Between any fixed probability levels η and $1 - \eta$, $0 < \eta < 1/2$, its width is $O_{k,\eta}(\sqrt{n\ell_k(n)})$. Chebyshev’s inequality also gives

$$\Pr(|H_n - \mathbb{E}H_n| \geq t\sqrt{n\ell_k(n)}) \leq C_k t^{-2} \quad (t > 0).$$

This is concentration about the finite-size expectation. The limiting threshold α_k of [16, Theorem 1.1] identifies $\lim_n \mathbb{E}H_n/n$, but supplies no bound on the bias $\mathbb{E}H_n - \alpha_k n$ at the fluctuation scale above. For $k = 3$, the separate companion [15, Theorem 1.1] proves that the limiting threshold α_3 is a computable real.

1.1 Earlier work and the new estimate

The location and the width of a threshold are different questions. Friedgut [12, Theorem 1.3] proved a sharp threshold sequence for each fixed clause size, without asserting convergence of that sequence. Achlioptas and Peres [2, Theorems 1–2] used a weighted second moment to locate the threshold within $O(k)$ of $2^k \log 2$ as k grows. Ding, Sly, and Sun [7, Theorem 1] proved the predicted limiting threshold for all sufficiently large k . Carenini’s paper [6, Corollary 1.2], made public on October 5, 2026, establishes the satisfiability conjecture for every fixed $k \geq 3$. We credit Carenini with priority for the resolution of the satisfiability conjecture. The companion [16, Theorem 1.1] establishes a limiting location for every fixed $k \geq 3$; its independent forcing argument gives the variance bound $O_k(n^{1+2/k})$ for the capped last-satisfiable prefix $\min(H_n - 1, 2^{k+1}n)$ [16, Proposition 3.3].

For transition width, Wilson [18, Corollary 4] proved an $\Omega_k(\sqrt{n})$ separation of central quantiles in the independent proper-clause model. We convert that separation into the variance lower bound, keeping the integer endpoints so the argument allows atoms. Carenini [5, Corollaries 7.10–7.11] proves an $O_{k,\eta}(n/\log n)$ window upper bound for fixed $k \geq 3$, including proper independent clauses. Her subsequent paper gives the polynomial window bound $O_{k,\eta}(n^{1/2+1/k})$ and the hitting-time variance bound $O_k(n^{1+2/k})$ [6, Theorems 1.1 and 1.3]; the window estimate, together with the Abbe–Montanari criterion, yields the limiting-threshold theorem credited above. The upper bounds here sharpen these fluctuation estimates. For $k = 3$, the companion [17] further removes the logarithmic loss in this paper’s upper bound.

Our upper bound uses the coordinate-omission form of the Efron–Stein inequality [4, 8]. Deleting one clause leaves its k variables available for root-flip tests of the remaining solutions. When no other clause meets two of these variables, the k tests are independent and use clauses of length $k - 1$; we account separately for such collisions. A set of assignments that is easily killed by those shorter clauses must also have a short expected lifetime under k -clauses. The replacement of a shorter constraint by a block of native clauses has a qualitative predecessor in Friedgut’s half-cube argument [12, Lemma 5.7]; sequential conditioning on a residual assignment set is developed in Carenini [5, Theorem 4.3 and Lemma 6.1]. We prove a block estimate uniform in the number of variables and the block length. It also covers assignment sets that no single k -clause can kill.

The main improvement comes after this replacement step. Instead of bounding the number of clauses incident to the deleted clause's variables by a logarithmic cutoff, we retain that number as a random parameter. Its fixed moments are bounded. The proof must therefore use the incidence count without destroying the independence of the shorter-clause tests or of future ordinary clauses. Conditioning on the incidence pattern through the cap achieves both. Collisions involving two of the deleted clause's variables are paid for by joint incidence and collision moments.

1.2 The route through the upper bound

Write $L = \lfloor Bn \rfloor$. If deleting C_i while preserving the other clause indices delays the capped first failure by D_i , then

$$\text{Var}(T_{n,B}) \leq \sum_{i=1}^L \mathbb{E} D_i^2.$$

The delay D_i counts the prefixes at which restoring C_i destroys satisfiability, so D_i^2 counts ordered pairs of such prefixes. The problem is to control this occupation time.

Here is the shorter-clause estimate used for that purpose. A collection of clauses *kills* $S \subseteq \{0,1\}^u$ if no member of S satisfies every clause. Let $q_d(S)$ be the probability that d independent proper $(k-1)$ -clauses kill S , and let $\tau_k(S)$ be the expected number of independent proper k -clauses through its first kill. For $u \geq k$, $d \geq 1$, and $\beta = k/(k-1)$, we prove in Section 2 that

$$\tau_k(S) q_d(S)^\beta \leq C_k d^\beta.$$

This estimate holds for every assignment set, with no lower cutoff on $q_d(S)$.

Section 3 connects this statement to clause deletion. Call the variables of C_i its roots, and fix them to the unique values that falsify C_i . Let S_m be the nonroot parts of solutions of the prefix with C_i omitted and the roots fixed this way. A clause disjoint from the roots is an ordinary k -clause on the nonroots. A clause meeting one root and satisfied by that root hides a $(k-1)$ -clause: its nonroot part becomes a constraint when that root is flipped. If restoring C_i destroys satisfiability, each allowed root flip must fail. In the absence of collisions this gives k independent tests killing S_m .

Conditional on the exposed incidence pattern, let Z be the number of clauses meeting the roots and put $d = \max\{1, Z\}$. The lifetime estimate bounds both the expected remaining duration of a deletion event and the total time spent at levels $q_d(S_m) \geq a$. Write $q = q_d(S_m)$. The k independent tests contribute q^k , while the expected remaining duration costs a factor $q^{-\beta}$. Integrating the resulting $q^{k-\beta}$ over time costs a second lifetime factor: the expected occupation above level a is bounded by a constant times $a^{-\beta}$. The near-zero part of the resulting integral is therefore

$$\int a^{k-2\beta-1} da.$$

At $k=3$, $k=2\beta$ and the finite cap produces a logarithm; for $k \geq 4$, $k > 2\beta$ and the integral is bounded. Section 4 then averages the root counts and proves the capped upper bound. Exponential tails remove a sufficiently late cap. Wilson's quantile bound supplies the separate lower-bound argument.

Section 5 removes a sufficiently late cap and records the exact first-unsatisfiable and last-satisfiable endpoint identities. Section 6 proves the lower bound, completing the variance theorem. Section 7 specializes the cap constants to three clauses and transfers the proper-clause concentration to the Poisson clause law used in the threshold companion. After that main

argument, Appendix A determines the limits of the two estimates for arbitrary assignment sets. Appendix B compares proper-clause models at different sizes, and Appendix C gives a second three-clause proof based on prefix exposure and incidence truncation.

2 Survival of a set of assignments

The deletion argument will leave a set of assignments that must pass random tests of length $k - 1$. We need to relate the probability that those tests eliminate the set to its lifetime under clauses of length k . This section proves the relation for every assignment set. Replacing a shorter constraint by a block of clauses of the original length has a qualitative predecessor in Friedgut's half-cube argument [12, Lemma 5.7].

Fix integers $k \geq 3$ and $u \geq k$, and put

$$r = k - 1, \quad \beta = \frac{k}{k - 1}.$$

A proper s -clause, for $1 \leq s \leq u$, chooses an s -element subset of the u variables uniformly and then chooses independent fair signs on that subset. Clauses are independent, so a whole clause may occur more than once. A collection of clauses *kills* $S \subseteq \{0, 1\}^u$ when no member of S satisfies every clause in the collection. For nonnegative integers d, g , define

$$q_d(S) = \mathbb{P}(d \text{ proper } r\text{-clauses kill } S), \quad R_g(S) = \mathbb{P}(g \text{ proper } k\text{-clauses kill } S).$$

For a nonempty set, $q_0(S) = R_0(S) = 0$. For the empty set these probabilities are one, including when there are zero clauses. Let $H_k(S)$ be the first killing index in an infinite sequence of proper k -clauses, counting the killing clause, and put

$$H_k(\emptyset) = 0, \quad \tau_k(S) = \mathbb{E}H_k(S).$$

These expectations are finite: for every integer $m \geq 0$,

$$\mathbb{P}(H_k(S) > m) \leq |S|(1 - 2^{-k})^m \leq 2^u \exp(-2^{-k}m). \quad (2.1)$$

Indeed, each fixed assignment survives each independent clause with probability $1 - 2^{-k}$, and the first inequality is a union bound.

2.1 Replacing one shorter clause

For a nonempty set S , a coordinate is *frozen* if every member of S has the same value there. If b coordinates are frozen, the probability that one proper s -clause kills S is

$$h_s(S) = \frac{(b)_s}{2^s(u)_s}, \quad 1 \leq s \leq u, \quad (2.2)$$

where $(a)_s = a(a - 1) \cdots (a - s + 1)$, with value zero for a nonnegative integer $a < s$. To kill S , every literal of the clause must be false for every assignment in S . Its variables must therefore be frozen, and each sign must oppose their common values. This characterizes the killing clauses and proves the formula.

Write $h = h_r(S)$. Then $0 \leq h \leq 2^{-r} \leq 1/4$. If $b \geq k$,

$$h_k(S) = h \frac{b - r}{2(u - r)} \geq \frac{1}{k} h^\beta. \quad (2.3)$$

In fact $b - r \geq b/k$, $u - r \leq u$, and

$$h = \frac{1}{2^r} \prod_{j=0}^{r-1} \frac{b-j}{u-j} \leq \left(\frac{b}{2u} \right)^r.$$

Substituting these bounds into the equality proves (2.3). Its hypothesis matters. For the cylinder whose first r coordinates are fixed to zero,

$$h_k(S) = 0, \quad h_r(S) = \frac{1}{2^r \binom{u}{r}} > 0. \quad (2.4)$$

No single k -clause kills this set. A sufficiently long block can still do so through the joint effect of its clauses.

Lemma 2.1 (One-clause replacement). *For an integer $k \geq 3$, let*

$$K_k = (2^k(k-1))^{k-1}.$$

For every integer $u \geq k$, every $S \subseteq \{0,1\}^u$, and every integer $g \geq 1$,

$$R_g(S) \geq q_1(S) - K_k g^{-(k-1)}. \quad (2.5)$$

Proof. For $S = \emptyset$, both probabilities are one. Suppose S is nonempty and retain b, r, h above. A block kills S if any one of its clauses does. Independence gives

$$R_g(S) \geq 1 - (1 - h_k(S))^g \geq 1 - e^{-gh_k(S)}. \quad (2.6)$$

First assume $b \geq k$. If $(g/k)h^{1/r} \geq 2$, then (2.3) gives $gh_k(S) \geq 2h$, and

$$R_g(S) \geq 1 - e^{-2h} \geq h.$$

For the last inequality, $1 - e^{-2h} - h$ is zero at $h = 0$ and has nonnegative derivative on $[0, 1/4]$. If instead $(g/k)h^{1/r} < 2$, then

$$h < (2k)^r g^{-r} \leq K_k g^{-r},$$

because $2k \leq 2^k(k-1)$. Nonnegativity of $R_g(S)$ now suffices.

If $b < r$, then $h = 0$, which also proves the inequality. The only remaining case is $b = r$. Formula (2.2) gives

$$h = \frac{1}{2^r \binom{u}{r}} \leq \left(\frac{r}{2u} \right)^r.$$

When $g < 2^{k+1}u$, this is less than $(2^k r/g)^r = K_k g^{-r}$. When $g \geq 2^{k+1}u$, apply (2.1) to the full cube:

$$R_g(S) \geq 1 - \exp((\log 2 - 2)u) \geq 1 - e^{-u} > \frac{1}{4} \geq h.$$

Here $u \geq k \geq 3$. This last bound uses the whole block and remains valid even though $h_k(S) = 0$. $\square$

2.2 Several tests and the expected lifetime

Replacing tests one at a time requires a bound for arbitrary residual sets: conditioning on the other clauses need not leave a set with any special description. The sequential conditioning follows the replacement argument of Carenini [5, Theorem 4.3]. The uniform additive block estimate in Lemma 2.1 takes the place of the cutoff-dependent comparison of single-clause killing probabilities in [5, Lemma 6.1]; the exceptional case $b = k - 1$ above rules out a cutoff-free positive power lower bound for h_k in terms of h_r .

Lemma 2.2 (Sequential replacement). *For integers $k \geq 3$ and $u \geq k$, every $S \subseteq \{0, 1\}^u$, and all integers $d, g \geq 1$,*

$$R_{dg}(S) \geq q_d(S) - dK_k g^{-(k-1)}. \quad (2.7)$$

Proof. Choose independent proper r -clauses $Q_1, \dots, Q_d$ and independent blocks $G_1, \dots, G_d$, each containing g proper k -clauses; take all these choices mutually independent. Let p_j be the probability that $G_1, \dots, G_j, Q_{j+1}, \dots, Q_d$ kill S , for $0 \leq j \leq d$.

For the replacement at index j , condition on $G_1, \dots, G_{j-1}, Q_{j+1}, \dots, Q_d$, and let S' be the members of S satisfying those clauses. The omitted Q_j and fresh G_j remain independent with their original laws. Their conditional killing probabilities are $q_1(S')$ and $R_g(S')$. Lemma 2.1 applies also if $S' = \emptyset$, so $p_j \geq p_{j-1} - K_k g^{-r}$ after averaging. Summing from $j = 1$ to d , using $p_0 = q_d(S)$ and $p_d = R_{dg}(S)$, proves the claim. $\square$

Proposition 2.3 (Lifetime from shorter-clause killing). *For an integer $k \geq 3$, put*

$$\beta = \frac{k}{k-1}, \quad K_k = (2^k(k-1))^{k-1}, \quad C_k = 4(2K_k)^{1/(k-1)}.$$

For every integer $u \geq k$, every $S \subseteq \{0, 1\}^u$, and every integer $d \geq 1$,

$$\tau_k(S) q_d(S)^\beta \leq C_k d^\beta. \quad (2.8)$$

Consequently, for nonempty S and $0 < a \leq q_d(S)$,

$$\tau_k(S) \leq C_k d^\beta a^{-\beta}. \quad (2.9)$$

There is no upper restriction on d and no positive lower cutoff on a .

Proof. For the empty set the lifetime is zero. If $S \neq \emptyset$ and $q_d(S) = 0$, the left side of (2.8) is zero, since the lifetime is finite. Otherwise put $q = q_d(S) > 0$ and choose

$$g = \left\lceil \left(\frac{2K_k d}{q} \right)^{1/(k-1)} \right\rceil.$$

Sequential replacement gives $R_{dg}(S) \geq q/2$. Split an infinite independent sequence of k -clauses into blocks of length dg . The events that these blocks kill the *original* set S are independent, each of probability at least $q/2$. The first successful block has expected index at most $2/q$, and the cumulative sequence kills S no later than the end of that block. Hence

$$\tau_k(S) \leq \frac{2dg}{q}.$$

The quantity inside the ceiling is at least one, so g is at most twice that quantity. It follows that

$$\tau_k(S) \leq 4(2K_k)^{1/(k-1)} \left(\frac{d}{q}\right)^{1+1/(k-1)} = C_k d^\beta q^{-\beta}.$$

This proves the product bound and then the stated level bound. $\square$

These inequalities also hold conditionally for a set S measurable with respect to an exposed history, provided the comparison clauses remain independent of that history. Apply the deterministic-set inequality at each realization of S . In particular, sequential replacement conditions only on other independent clause positions; it does not condition the fresh clauses on a satisfiability event.

3 Clause omission and root tests

We now turn the lifetime estimate into a bound for the effect of deleting one clause. Its variables provide tests of the assignments that remain after deletion. We expose the incidence pattern through the whole cap, so that the number of tests is known, while keeping both their contents and the future ordinary clauses independent. The tests bound the probability of a deletion event; the ordinary clauses bound its remaining duration.

Fix an integer $k \geq 3$, a real $B > 0$, and $n \geq 2k$. Put $L = \lfloor Bn \rfloor$, and for now suppose $L \geq 1$. Let $C_1, C_2, \dots$ be independent uniform proper k -clauses on n variables, let H be the first unsatisfiable prefix index, and put $T = \min\{H, L\}$. For $1 \leq i \leq L$, define

$$B_m^{(i)} = \bigwedge_{\substack{1 \leq j \leq m \\ j \neq i}} C_j, \quad T^{(i)} = \min\{\inf\{m \geq 0 : B_m^{(i)} \notin \text{SAT}\}, L\}, \quad D_i = T^{(i)} - T.$$

The infimum is over integers and $\inf \emptyset = \infty$. Deleting C_i retains all other clause indices. It can only delay unsatisfiability, so $D_i \geq 0$; moreover $T^{(i)}$ depends only on the first L clauses other than C_i .

Lemma 3.1 (Coordinate omission). *For the independent clause process above,*

$$\text{Var}(T) \leq \sum_{i=1}^L \mathbb{E} D_i^2. \quad (3.1)$$

Proof. This is the coordinate-omission form of the Efron–Stein method [8], recorded in [4, equation (3.6)]. Here is a direct martingale proof. Let $\mathcal{H}_j = \sigma(C_1, \dots, C_j)$, with $\mathcal{H}_0$ trivial, and write

$$M_i = \mathbb{E}[T \mid \mathcal{H}_i] - \mathbb{E}[T \mid \mathcal{H}_{i-1}].$$

Since $T^{(i)}$ omits the independent coordinate C_i , $\mathbb{E}[T^{(i)} \mid \mathcal{H}_i] = \mathbb{E}[T^{(i)} \mid \mathcal{H}_{i-1}]$. Thus, with $X_i = T - T^{(i)}$,

$$M_i = \mathbb{E}[X_i \mid \mathcal{H}_i] - \mathbb{E}[X_i \mid \mathcal{H}_{i-1}].$$

The tower property gives

$$\mathbb{E} M_i^2 = \mathbb{E}(\mathbb{E}[X_i \mid \mathcal{H}_i])^2 - \mathbb{E}(\mathbb{E}[X_i \mid \mathcal{H}_{i-1}])^2 \leq \mathbb{E} X_i^2 = \mathbb{E} D_i^2,$$

where the inequality uses conditional Jensen. The martingale differences are orthogonal and T is $\mathcal{H}_L$ -measurable, so summing proves the claim. $\square$

3.1 What the two filtrations reveal

Fix i and condition on a value of C_i . Until the end of Section 4, probabilities and expectations refer to this conditional law. The bounds will be uniform in the fixed clause. Its set R of k variables will be called the roots, and $t \in \{0, 1\}^R$ is their unique assignment falsifying C_i . There are $u = n - k \geq k$ remaining variables. Write $B_m = B_m^{(i)}$.

For $i \leq m < L$, define the deletion event

$$A_m = \{B_m \in \text{SAT}, B_m \wedge C_i \notin \text{SAT}\} = \{T \leq m < T^{(i)}\}. \quad (3.2)$$

On this event every solution of B_m has root values t . The two prefix processes agree before i , and hence

$$D_i = \sum_{m=i}^{L-1} \mathbf{1}_{A_m}. \quad (3.3)$$

In particular $D_L = 0$: the displayed sum is then empty. The upper endpoint is $L - 1$ because both failure times have already been stopped at L .

For $i \leq m \leq L$, define $S_m \subseteq \{0, 1\}^u$ to be the nonroot restrictions of solutions of B_m with root values t . The sets are defined on every outcome and decrease with m ; for $m < L$, A_m implies that S_m is nonempty. We need to reveal this set while retaining independent randomness to test whether one of its assignments allows a root to be flipped.

The *root type* of a clause records the subset of its variables in R and the signs of their literals. A clause is *ordinary* if that subset is empty. It is a *test clause* if the subset contains exactly one root and its literal is true at t . Let O and Q be the ordinary and test indices among $\{1, \dots, L\} \setminus \{i\}$.

Reveal the root type at every such index through L . At indices in $\{1, \dots, L\} \setminus (\{i\} \cup O \cup Q)$, also reveal the complete signed nonroot part. Let $\mathcal{E}$ be the sigma-field generated by this static information. Thus O, Q and the counts

$$\begin{aligned} Z &= \#\{j \leq L : j \neq i, C_j \text{ meets } R\}, \\ V &= \#\{j \leq L : j \neq i, C_j \text{ meets at least two roots}\}, \quad d = \max\{1, Z\} \end{aligned} \quad (3.4)$$

are $\mathcal{E}$ -measurable. We call a clause counted by V a *collision*. The ordinary clauses and the nonroot parts of test clauses have not yet been revealed.

Lemma 3.2 (Conditional product law). *Conditional on $\mathcal{E}$, the clauses at indices in O are independent uniform proper k -clauses on the u nonroots. The nonroot parts at indices in Q are independent uniform proper $(k - 1)$ -clauses on the nonroots. These two collections are mutually independent.*

Proof. For a prescribed root type with h roots and prescribed root signs, there are exactly $2^{k-h} \binom{u}{k-h}$ possible signed nonroot parts. Since complete proper k -clauses are equiprobable, these parts are equiprobable conditional on the type. Original clause independence makes this conditional law a product over the indices. Once the types are fixed, O and Q are fixed index sets. Revealing the parts outside $O \cup Q$ preserves the product law on the other indices. Their residual lengths are k on O and $k - 1$ on Q , as asserted. $\square$

A test clause is already true under t , so its nonroot part imposes no condition on S_m . Consequently S_m is measurable with respect to

$$\mathcal{G}_m = \sigma(\mathcal{E}, C_j : j \in O, j \leq m).$$

We will bound the probability of A_m under this filtration, which knows S_m but leaves the test parts hidden. To bound the remaining duration on A_m , we also need to know whether A_m has occurred. For that purpose, enlarge the filtration by revealing the past test parts:

$$\mathcal{F}_m^* = \sigma(\mathcal{G}_m, \text{nonroot parts of } C_j : j \in Q, j \leq m).$$

The full baseline prefix B_m , and thus A_m , is $\mathcal{F}_m^*$ -measurable. Under $\mathcal{G}_m$, the past test parts remain independent with their original conditional laws. Under either filtration, the future ordinary clauses remain independent uniform proper k -clauses. Both assertions follow from Lemma 3.2. The static information includes future nonordinary data, but it includes no future ordinary contents.

3.2 Flipping a root gives an independent test

Recall that $q_d(S)$ is the probability that d independent proper $(k-1)$ -clauses kill S . Here $d = \max\{1, Z\}$ is already known under $\mathcal{E}$. Put

$$\beta = \frac{k}{k-1}, \quad q_m = q_d(S_m).$$

For $j \in R$, let $\mathcal{I}_{j,m}$ be the test indices through m whose root is j , and put $s_{j,m} = |\mathcal{I}_{j,m}|$. Then $s_{j,m} \leq Z \leq d$. Let $J_m \subseteq R$ consist of roots j for which no collision through m has exactly one root literal true under t , at root j . A collision can exclude at most one root, so

$$|J_m| \geq k - V. \tag{3.5}$$

The index sets and counts in this paragraph are $\mathcal{E}$ -measurable.

Lemma 3.3 (Root-flip tests). *For $i \leq m < L$, on the event $\{|J_m| \geq 1\}$,*

$$\mathbb{P}(A_m \mid \mathcal{G}_m) \leq \mathbf{1}_{\{S_m \neq \emptyset\}} q_m^{|J_m|}. \tag{3.6}$$

Proof. Suppose A_m occurs, and fix $j \in J_m$. The nonroot parts at the indices in $\mathcal{I}_{j,m}$ must kill S_m . If they did not, choose $x \in S_m$ satisfying all of them. Extend x by the root assignment t , then flip root j . We check the clauses of B_m .

A clause with no true root literal under t has a satisfied nonroot part, because (t, x) satisfies B_m . The flip does not change that part. A clause with a true literal on another root remains true. Every remaining clause has j as its unique true root under t . It cannot be a collision, by $j \in J_m$, so it is a test clause at an index in $\mathcal{I}_{j,m}$; its nonroot part is satisfied by the choice of x . The flipped assignment therefore satisfies B_m with root values different from t , contradicting A_m .

Conditional on $\mathcal{G}_m$, the test parts belonging to distinct roots are disjoint independent families of proper $(k-1)$ -clauses. For nonempty S_m , a family of $s_{j,m} \leq d$ tests kills it with probability $q_{s_{j,m}}(S_m) \leq q_d(S_m)$. This also covers $s_{j,m} = 0$, whose killing probability is zero. The necessary tests for all roots in J_m thus have conditional probability at most $q_m^{|J_m|}$. Finally A_m requires $S_m \neq \emptyset$. $\square$

The proof has kept the actual incidence count d , rather than discarding environments where it exceeds a chosen cutoff. We next bound duration with that same parameter; its moments will be averaged only after all conditional estimates have been proved.

3.3 From ordinary clauses to chronological duration

For this comparison, keep the actual schedule and ordinary clauses through L . At every index after L , append an auxiliary independent uniform proper k -clause on the u nonroots. The continuation is independent of the entire original process.

For $i \leq m < L$ and $S_m \neq \emptyset$, let K_m be the number of ordinary clauses strictly after m , including the first one that completes the killing of the original set S_m . Use the continuation if necessary. Let U_m be the number of chronological indices after m through that killing index. Set $K_m = U_m = 0$ if $S_m = \emptyset$. The conditional product law and the independent continuation give

$$\mathbb{E}[K_m \mid \mathcal{F}_m^*] = \mathbb{E}[K_m \mid \mathcal{G}_m] = \tau_k(S_m). \quad (3.7)$$

There are at most Z nonordinary indices between m and the kill: every one occurs before or at L , and the deleted index i is not after m . Thus, pathwise,

$$U_m \leq K_m + Z, \quad \mathbb{E}[U_m \mid \mathcal{F}_m^*] \leq \tau_k(S_m) + Z, \quad \mathbb{E}[U_m \mid \mathcal{G}_m] \leq \tau_k(S_m) + Z. \quad (3.8)$$

The continuation makes these statements valid even when no pre-cap ordinary clause completes the kill.

Lemma 3.4 (Weighted deletion delay). *Let*

$$W_m = \min\{L, \tau_k(S_m) + Z\}.$$

This is $\mathcal{G}_m$ -measurable. For $i \leq m < L$, on A_m ,

$$\mathbb{E} \left[\sum_{\ell=m+1}^{L-1} \mathbf{1}_{A_\ell} \middle| \mathcal{F}_m^* \right] \leq W_m. \quad (3.9)$$

Consequently

$$\mathbb{E}[D_i^2 \mid \mathcal{E}] \leq 3 \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{A_m} W_m \mid \mathcal{E}]. \quad (3.10)$$

Proof. On A_m , every later solution of B_ℓ must have root values t and a nonroot restriction in S_m . Once the future ordinary clauses kill that original set, there can be no such solution, including at the killing index. If the kill is after L , the remaining capped interval ends first. In either case the number of later deletion events is at most U_m , and is also at most L . Taking conditional expectations and using (3.8) gives (3.9).

On A_m , the set S_m is nonempty and therefore needs at least one ordinary clause to be killed. Hence $\tau_k(S_m) \geq 1$ and $W_m \geq 1$. Expand the square of (3.3). For each cross term condition on the earlier $\mathcal{F}_m^*$; it contains A_m . The tower property gives

$$\begin{aligned} \mathbb{E}[D_i^2 \mid \mathcal{E}] &= \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{A_m} \mid \mathcal{E}] \\ &\quad + 2 \sum_{m=i}^{L-1} \mathbb{E} \left[\mathbf{1}_{A_m} \mathbb{E} \left[\sum_{\ell=m+1}^{L-1} \mathbf{1}_{A_\ell} \middle| \mathcal{F}_m^* \right] \middle| \mathcal{E} \right] \\ &\leq \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{A_m} (1 + 2W_m) \mid \mathcal{E}] \\ &\leq 3 \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{A_m} W_m \mid \mathcal{E}]. \end{aligned} \quad \square$$

Both the root-test probability and the remaining duration are now controlled by the same set S_m . We still have to bound how long that set can spend at each level of $q_d(S_m)$, and then average the incidence counts.

4 Occupation and incidence moments

Keep the deleted index i , its fixed clause C_i , and the environment $\mathcal{E}$ of Section 3. We first bound the time spent at each level of the shorter-clause killing probability $q_m = q_d(S_m)$. We then combine that occupation bound with the root tests and average the incidence counts.

In this section C_k denotes a positive constant depending only on the fixed k , which may increase between occurrences. It need not equal the explicit constant in Proposition 2.3. Recall $\beta = k/(k-1)$, $d = \max\{1, Z\}$, and $W_m = \min\{L, \tau_k(S_m) + Z\}$. On $S_m \neq \emptyset$,

$$W_m q_m^\beta \leq C_k d^\beta. \quad (4.1)$$

Indeed, Proposition 2.3 bounds $\tau_k(S_m) q_m^\beta$ by a constant times d^β , and

$$Z q_m^\beta \leq Z \leq d \leq d^\beta.$$

The product bound remains valid when $q_m = 0$.

4.1 Time spent above a level

Lemma 4.1 (Conditional occupation). *There is a constant $C_k \geq 1$, independent of the environment, such that for every $0 < a \leq 1$, almost surely,*

$$\sum_{m=i}^{L-1} \mathbb{P}(S_m \neq \emptyset, q_m \geq a \mid \mathcal{E}) \leq \min\{L, C_k d^\beta a^{-\beta}\}. \quad (4.2)$$

Proof. Define

$$N_a = \sum_{m=i}^{L-1} \mathbf{1}_{\{S_m \neq \emptyset, q_m \geq a\}},$$

and let σ_a be the first index counted by N_a , or ∞ if there is none. Because d is known in $\mathcal{E}$ and S_m is $\mathcal{G}_m$ -measurable, $\{\sigma_a = m\} \in \mathcal{G}_m$.

On $\{\sigma_a = m\}$, no eligible index precedes m . The future ordinary clauses used to define U_m kill the original S_m at index $m + U_m$. If that index is at most L , then S_ℓ is empty there and thereafter: a member would have to lie in S_m and satisfy those same ordinary clauses. The eligible indices are therefore among $m, \dots, m + U_m - 1$. If the kill is after L , the capped interval ends earlier. Thus in both cases

$$N_a \leq U_m \quad \text{on } \{\sigma_a = m\}.$$

The first eligible index is included in these U_m possible indices.

On $\{\sigma_a = m\}$, the lifetime bound and (3.8) give

$$\mathbb{E}[U_m \mid \mathcal{G}_m] \leq \tau_k(S_m) + Z \leq C_k d^\beta a^{-\beta},$$

after increasing C_k , since $q_m \geq a$ and $Z \leq d \leq d^\beta a^{-\beta}$. The first-visit events are disjoint, so the tower property yields

$$\begin{aligned}\mathbb{E}[N_a \mid \mathcal{E}] &\leq \sum_{m=i}^{L-1} \mathbb{E} \left[\mathbf{1}_{\{\sigma_a=m\}} \mathbb{E}[U_m \mid \mathcal{G}_m] \mid \mathcal{E} \right] \\ &\leq C_k d^\beta a^{-\beta} \sum_{m=i}^{L-1} \mathbb{P}(\sigma_a = m \mid \mathcal{E}) \leq C_k d^\beta a^{-\beta}.\end{aligned}$$

Also $N_a \leq L$. These two bounds prove the claim. $\square$

Only the first visit to the level is stopped on. All later eligible indices are charged to the remaining lifetime of the same nested set. This is why the argument needs fresh future ordinary contents under $\mathcal{G}_m$, even though $\mathcal{E}$ has already exposed future nonordinary data.

To apply the occupation bound to the squared deletion delay, first consider an environment with $V = 0$. All k root tests are then available. Lemma 3.3 supplies the factor q_m^k in the weighted-delay bound (3.10). On a nonempty S_m , (4.1) gives

$$W_m q_m^k = (W_m q_m^\beta) q_m^{k-\beta} \leq C_k d^\beta q_m^{k-\beta}.$$

The remaining occupation sum has exponent

$$\gamma = k - \beta = \frac{k(k-2)}{k-1}.$$

The lifetime estimate has controlled the remaining deletion duration; Lemma 4.1 contributes a second lifetime factor $a^{-\beta}$ by bounding the time spent above level a . The resulting layer-cake integral contains $a^{\gamma-\beta-1}$. Its integrability at zero is determined by $\gamma - \beta = k(k-3)/(k-1)$, which is zero exactly at $k = 3$.

Lemma 4.2 (Integrated occupation). *For $L \geq 1$, almost surely and with a constant independent of the environment,*

$$\sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{\{S_m \neq \emptyset\}} q_m^\gamma \mid \mathcal{E}] \leq \begin{cases} C_k d^\beta (1 + \log L), & k = 3, \\ C_k d^\beta, & k \geq 4. \end{cases} \quad (4.3)$$

Proof. Since $0 \leq q_m \leq 1$, the layer-cake identity and Lemma 4.1 give

$$\begin{aligned}&\sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{\{S_m \neq \emptyset\}} q_m^\gamma \mid \mathcal{E}] \\ &= \int_0^1 \gamma a^{\gamma-1} \sum_{m=i}^{L-1} \mathbb{P}(S_m \neq \emptyset, q_m \geq a \mid \mathcal{E}) da \\ &\leq \int_0^1 \gamma a^{\gamma-1} \min\{L, A a^{-\beta}\} da, \quad A = C_k d^\beta \geq 1. \end{aligned} \quad (4.4)$$

The terms are nonnegative, so the interchange of conditional expectation, the finite sum, and integration is valid.

For $k \geq 4$, $\gamma > \beta$. Bounding the minimum by $A a^{-\beta}$ gives

$$\int_0^1 \gamma a^{\gamma-1} \min\{L, A a^{-\beta}\} da \leq \frac{\gamma A}{\gamma - \beta}.$$

This is a constant times A for fixed k .

For $k = 3$, $\gamma = \beta = 3/2$. If $A \geq L$, the integral is at most $L \leq A$. If $A < L$, split at $a_* = (A/L)^{1/\beta}$. The lower part equals $La_*^\beta = A$; the upper part is

$$\beta A \int_{a_*}^1 \frac{da}{a} = A \log(L/A).$$

Thus the integral is at most $A[1 + \log_+(L/A)] \leq A(1 + \log L)$, where $\log_+ x = \max\{\log x, 0\}$. This proves both cases. $\square$

4.2 The cost of collisions

The number V of collisions through the whole cap is fixed under $\mathcal{E}$. With no collisions all k root tests are available. One collision may remove one test; two or more collisions will be handled by the deterministic cap.

Proposition 4.3 (Conditional squared delay). *For $L \geq 1$, set*

$$h_k(L) = \begin{cases} 1 + \log L, & k = 3, \\ 1, & k \geq 4. \end{cases}$$

Then, almost surely,

$$\mathbb{E}[D_i^2 \mid \mathcal{E}] \leq C_k d^{2\beta} h_k(L) \mathbf{1}_{\{V=0\}} + C_k L d^\beta \mathbf{1}_{\{V=1\}} + L^2 \mathbf{1}_{\{V \geq 2\}}. \quad (4.5)$$

Proof. Because W_m is $\mathcal{G}_m$ -measurable, the tower property and the root-flip test bound imply, for every integer $1 \leq j \leq k$ on the $\mathcal{E}$ -measurable event $\{|J_m| \geq j\}$,

$$\mathbb{E}[\mathbf{1}_{A_m} W_m \mid \mathcal{E}] \leq \mathbb{E}[\mathbf{1}_{\{S_m \neq \emptyset\}} W_m q_m^j \mid \mathcal{E}].$$

Here we used $q_m \leq 1$ if more than j tests are available.

If $V = 0$, take $j = k$ and use $W_m q_m^k \leq C_k d^\beta q_m^\gamma$ on nonempty sets, as above. Substituting in (3.10) and applying integrated occupation gives $C_k d^{2\beta} h_k(L)$.

If $V = 1$, at least $k - 1$ tests are available by (3.5). Since $k - 1 \geq \beta$ for $k \geq 3$,

$$\mathbf{1}_{\{S_m \neq \emptyset\}} W_m q_m^{k-1} \leq C_k d^\beta.$$

Use this bound at each of the at most L indices in (3.10). It gives $C_k L d^\beta$. Finally, when $V \geq 2$, the pathwise bound $0 \leq D_i \leq L$ gives $D_i^2 \leq L^2$. This includes environments with no available test. $\square$

The one-collision contribution contains d^β . Its average therefore needs the joint distribution of incidences and collisions, not just an upper bound on the probability of a collision.

4.3 Joint moments of incidences and collisions

Lemma 4.4 (Weighted root counts). *Fix integers $k \geq 3$ and $n \geq 2k$, a real $B > 0$, and a set of k roots. Take M independent uniform proper k -clauses on n variables, where M is a nonnegative integer with $M \leq Bn$. Let Z count clauses meeting a root, V count clauses meeting at least two roots, and $d = \max\{1, Z\}$. For every fixed real $p \geq 0$,*

$$\mathbb{E} d^p \leq C_{k,B,p}. \quad (4.6)$$

For every fixed positive integer r ,

$$\mathbb{E}[d^p \mathbf{1}_{\{V \geq r\}}] \leq C_{k,B,p,r} n^{-r}. \quad (4.7)$$

Proof. For one clause, let I indicate meeting a root and let J indicate meeting at least two. Write $p_1 = \mathbb{P}(I = 1)$ and $p_2 = \mathbb{P}(J = 1)$. Union bounds over roots and pairs of roots give

$$p_1 \leq \frac{k^2}{n}, \quad p_2 \leq \binom{k}{2} \frac{k(k-1)}{n(n-1)} \leq \frac{k^2(k-1)^2}{n^2}. \quad (4.8)$$

Thus $Z \sim \text{Bin}(M, p_1)$, and

$$\mathbb{E}e^Z = (1 + p_1(e-1))^M \leq \exp(Bk^2(e-1)).$$

For $K_p = \sup_{x \geq 0} \max\{1, x\}^p e^{-x} < \infty$, we have $d^p \leq K_p e^Z$. This proves (4.6).

For the collision bound, retain the dependence between Z and V . Write $(a)_r = a(a-1)\cdots(a-r+1)$. For every nonnegative function f on the nonnegative integers and $1 \leq r \leq M$,

$$\mathbb{E}[f(Z)(V)_r] = (M)_r p_2^r \mathbb{E}f(r+X), \quad X \sim \text{Bin}(M-r, p_1). \quad (4.9)$$

To prove the identity, expand $(V)_r$ over ordered r -tuples of distinct clause indices. A collision at each chosen index has probability p_2^r and forces its incidence indicator to be one. The remaining $M-r$ incidence indicators retain their independent Bernoulli laws with parameter p_1 . Each ordered tuple has the same expectation, giving the formula.

Since $\mathbf{1}_{\{V \geq r\}} \leq (V)_r / r!$, use $f(x) = \max\{1, x\}^p$ in (4.9):

$$\begin{aligned} \mathbb{E}[d^p \mathbf{1}_{\{V \geq r\}}] &\leq \frac{(M)_r p_2^r}{r!} \mathbb{E} \max\{1, r+X\}^p \\ &\leq \frac{(M)_r p_2^r}{r!} K_p e^r \exp(Bk^2(e-1)) \leq C_{k,B,p,r} n^{-r}. \end{aligned}$$

The last inequality uses $M \leq Bn$ and (4.8). If $r > M$, the left side is zero. The same definitions cover $M = 0$ and $p = 0$. $\square$

These moments are taken with the deleted clause fixed and the other clauses still random. Once $\mathcal{E}$ is fixed, Z and V are known; their rarity returns only when we average over $\mathcal{E}$.

4.4 Completing the capped upper bound

Proof of the capped upper bound in Theorem 1.1. Suppose first that $n \geq 2k$ and $L \geq 1$, and keep C_i fixed. The remaining $L-1 \leq Bn$ clauses satisfy Lemma 4.4. Averaging (4.5) over $\mathcal{E}$, and using the lemma with $p = 2\beta$ in (4.6) and $(p, r) = (\beta, 1), (0, 2)$ in (4.7), gives

$$\begin{aligned} \mathbb{E}[D_i^2 \mid C_i] &\leq C_k h_k(L) \mathbb{E}[d^{2\beta} \mid C_i] + C_k L \mathbb{E}[d^\beta \mathbf{1}_{\{V=1\}} \mid C_i] + L^2 \mathbb{P}(V \geq 2 \mid C_i) \\ &\leq C_{k,B} h_k(L) + C_{k,B} L n^{-1} + C_{k,B} L^2 n^{-2} \\ &\leq C_{k,B} h_k(L). \end{aligned}$$

The constants are uniform in i and in the value of C_i . Remove this conditioning and use coordinate omission:

$$\text{Var}(T) \leq \sum_{i=1}^L \mathbb{E} D_i^2 \leq C_{k,B} L h_k(L).$$

Since $L \leq Bn$, this is at most

$$C_{k,Bn} \ell_k(n), \quad \ell_k(n) = \begin{cases} \log(en), & k = 3, \\ 1, & k \geq 4. \end{cases}$$

For $k = 3$, we used $1 + \log L \leq C_B \log(en)$ when $L \geq 1$. If $L = 0$, the capped variable is identically zero. The finitely many remaining sizes $k \leq n < 2k$ are covered by increasing the constant, since $0 \leq T \leq \lfloor Bn \rfloor$. This proves the upper bound for every fixed positive B and every $n \geq k$. $\square$

5 Caps, uncapped times, and finite-size means

The capped upper bound uses only finitely many clause coordinates. We now choose a sufficiently late cap and remove it, compare the first-unsatisfiable and last-satisfiable conventions with their exact endpoint corrections, and identify the limiting normalized means. The lower estimate, which requires a different input, is proved in Section 6.

5.1 Exact cap comparisons

Fix $k \geq 3$, put $q_k = 1 - 2^{-k}$, and write $P_n(m) = \mathbb{P}(H_n > m)$. We also write $F_{n,m} = F_m$ when displaying the number of variables. Each assignment survives m independent proper clauses with probability q_k^m . The expected number of surviving assignments therefore gives

$$P_n(m) \leq \min\{1, 2^n q_k^m\} \quad (m \in \mathbb{Z}_{\geq 0}). \quad (5.1)$$

In particular, H_n is almost surely finite and has finite moments of every order. For integer caps $a \geq 1$ and $b \geq 0$, define

$$X_a = \min\{H_n, a\}, \quad V_b = \min\{H_n - 1, b\}.$$

These have the exact prefix identities

$$P_n(m) = \mathbb{P}(X_a > m) \quad (0 \leq m < a), \quad P_n(m) = \mathbb{P}(V_b \geq m) \quad (0 \leq m \leq b). \quad (5.2)$$

All clause counts in these identities are integers.

Proposition 5.1 (Finite and infinite cap comparisons). *For $b \geq a$, put $R_{a,b} = V_b - X_a + 1$ and $D_a = H_n - X_a$. Then*

$$R_{a,b} = \sum_{m=a}^b \mathbf{1}_{\{H_n > m\}}, \quad 0 \leq R_{a,b} \leq (b - a + 1) \mathbf{1}_{\{H_n > a\}}, \quad (5.3)$$

$$D_a = (H_n - a)_+. \quad (5.4)$$

For either $Z = R_{a,b}$ or $Z = D_a$, writing $\xi_{n,a} = 2^n q_k^a$,

$$\mathbb{E}Z \leq 2^k \xi_{n,a}, \quad \mathbb{E}Z^2 \leq (2^{2k+1} - 2^k) \xi_{n,a}. \quad (5.5)$$

If $X = V_b$ or $X = H_n$, respectively, then

$$0 \leq \sqrt{\text{Var}(X)} - \sqrt{\text{Var}(X_a)} \leq 2^k \sqrt{1 + q_k} \xi_{n,a}^{1/2}, \quad (5.6)$$

$$0 \leq \text{Var}(X) - \text{Var}(X_a) \leq [2a \cdot 2^k + 2^{2k+1} - 2^k] \xi_{n,a}. \quad (5.7)$$

The mean identities are $\mathbb{E}V_b = \mathbb{E}X_a - 1 + \mathbb{E}R_{a,b}$ and $\mathbb{E}H_n = \mathbb{E}X_a + \mathbb{E}D_a$.

Proof. The finite identity follows by considering $H_n \leq a$ and $H_n > a$; it also gives $R_{a,b} = \min\{(H_n - a)_+, b - a + 1\}$. For every integer $r \geq 0$,

$$\mathbb{P}(D_a > r) = P_n(a + r) \leq \xi_{n,a} q_k^r.$$

The integer tail-sum identities yield

$$\begin{aligned} \mathbb{E}D_a &= \sum_{r=0}^{\infty} P_n(a + r) \leq \frac{\xi_{n,a}}{1 - q_k}, \\ \mathbb{E}D_a^2 &= \sum_{r=0}^{\infty} (2r + 1)P_n(a + r) \leq \frac{1 + q_k}{(1 - q_k)^2} \xi_{n,a}. \end{aligned}$$

For $R_{a,b}$ the same sums end at $r = b - a$, so the bounds also apply. Since $X_a Z = aZ$ for either remainder,

$$\text{Cov}(X_a, Z) = (a - \mathbb{E}X_a)\mathbb{E}Z \geq 0.$$

The constant shift in the finite-cap identity does not affect variance. Thus

$$\text{Var}(X) - \text{Var}(X_a) = 2(a - \mathbb{E}X_a)\mathbb{E}Z + \text{Var}(Z) \geq 0.$$

Using (5.5) proves (5.7). The triangle inequality for centered variables in L^2 gives

$$|\sqrt{\text{Var}(X)} - \sqrt{\text{Var}(X_a)}| \leq \sqrt{\text{Var}(Z)} \leq \sqrt{\mathbb{E}Z^2},$$

which proves (5.6). □

Recall the first-moment cap density

$$U_k = \frac{\log 2}{-\log q_k}. \quad (5.8)$$

For any fixed $B > U_k$, let $\rho_{k,B} = 2q_k^B < 1$. The cap $a = \lfloor Bn \rfloor$ satisfies

$$\xi_{n,a} = q_k^{\lfloor Bn \rfloor - Bn} \rho_{k,B}^n \leq q_k^{-1} \rho_{k,B}^n. \quad (5.9)$$

The factor q_k^{-1} accounts for the floor; a ceiling cap needs no such factor. Equality $B = U_k$ does not give exponential decay by this argument. In particular, $B = 2^{k+1}$ is sufficient for every fixed $k \geq 3$.

Corollary 5.2 (Uncapped variance and finite-size concentration). *For independent uniform proper clauses with fixed $k \geq 3$,*

$$\text{Var}(H_n) = \begin{cases} O(n \log n), & k = 3, \\ O_k(n), & k \geq 4. \end{cases} \quad (5.10)$$

The same upper bounds hold for the uncapped last satisfiable index $H_n - 1$ and, for every fixed $B > 0$, for $\min\{H_n - 1, \lfloor Bn \rfloor\}$. Write $s_k(n) = \sqrt{n \ell_k(n)}$, with ℓ_k as in the introduction. For each of these statistics, or for $T_{n,B}$ from Theorem 1.1, there is a constant C (depending on fixed k and, when present, B) such that

$$\mathbb{P}(|X - \mathbb{E}X| \geq t s_k(n)) \leq C t^{-2} \quad (t > 0) \quad (5.11)$$

for all sufficiently large n .

Proof. Choose a fixed $B > U_k$ and use the capped upper bound of Theorem 1.1 for $X_a = T_{n,B}$, $a = \lfloor Bn \rfloor$. Equations (5.7) and (5.9) transfer its bound to H_n , with an additive $O_{k,B}(n\rho_{k,B}^n)$ error. Subtracting one leaves variance unchanged. For any real random variable Y with finite variance, any independent copy Y' , and any 1-Lipschitz function f ,

$$\text{Var}(f(Y)) = \frac{1}{2}\mathbb{E}(f(Y) - f(Y'))^2 \leq \frac{1}{2}\mathbb{E}(Y - Y')^2 = \text{Var}(Y).$$

Apply this to $f(y) = \min\{y, \lfloor Bn \rfloor\}$ and $Y = H_n - 1$. Finally, Chebyshev's inequality gives (5.11). Increasing the upper-bound constants covers the finitely many remaining $n \geq k$, since all uncapped moments are finite. $\square$

The mean correction is more precise than its normalized limiting effect: for $a = \lfloor Bn \rfloor$ with $B > U_k$,

$$0 \leq \frac{\mathbb{E}H_n - \mathbb{E}T_{n,B}}{n} \leq \frac{2^k q_k^{-1}}{n} \rho_{k,B}^n. \quad (5.12)$$

Theorem 1.1 of [16] gives a unique positive finite threshold α_k in exactly this proper-clause model. It also implies

$$\frac{\mathbb{E}H_n}{n} \longrightarrow \alpha_k, \quad \frac{\mathbb{E}T_{n,B}}{n} \longrightarrow \alpha_k \quad (B > U_k). \quad (5.13)$$

Here is the explicit expectation argument. Take $b = 2^{k+1}n$. The theorem and monotonicity imply $\min\{H_n - 1, b\}/n \rightarrow \alpha_k$ in probability; the first-moment bound implies $\alpha_k \leq U_k < 2^{k+1}$. The bounded statistic therefore converges in mean. Proposition 5.1 first compares it with X_b , then with H_n ; the errors after the one-index shift are exponentially small. Equation (5.12) handles the remaining caps. The same conclusion holds for last-satisfiable caps of density $B > U_k$. These limits give no rate for $\mathbb{E}H_n - \alpha_k n$ or for the corresponding capped bias. The concentration in (5.11) is centered at each finite-size expectation.

6 Quantile separation and the variance lower bound

The remaining part of Theorem 1.1 is a linear lower bound. We use a width theorem for the proper-clause model and convert it into a second-moment estimate. Keeping the two integer quantiles explicit makes the argument valid when the hitting-time law has atoms.

An elementary sparse-prefix estimate will also supply the positive lower bounds on finite-size means in Appendices B and C. We record it before turning to the quantile argument.

Lemma 6.1 (A sparse satisfiable prefix). *For every real $0 < c < e^{-2}$, every integer $k \geq 3$, and every $n \geq k$, put $j = \lfloor cn \rfloor$. In the independent proper-clause model,*

$$1 - P_n(j) \leq \frac{ec}{n^2} \sum_{t=1}^{\infty} t^2 (e^2 c)^t = O_c(n^{-2}). \quad (6.1)$$

Proof. Form the bipartite incidence graph between the j clause positions and the n variables. A matching covering all clause positions supplies a satisfying assignment: set each matched variable to satisfy its literal in the clause matched to it. This criterion and the deficient-set union bound below appear in Franco and Van Gelder [9, Section 8]. By Hall's theorem [14, Theorem 1], failure of such a matching implies that some set of t variables contains at least $t + 1$ whole clauses. A

proper k -clause lies in a specified t -set with probability $(t)_k/(n)_k \leq (t/n)^k$, interpreted as zero when $t < k$. Thus, using $k \geq 3$, $j \leq cn$, and $\binom{a}{b} \leq (ea/b)^b$,

$$\begin{aligned} 1 - P_n(j) &\leq \sum_{\substack{1 \leq t \leq n \\ t+1 \leq j}} \binom{n}{t} \binom{j}{t+1} (t/n)^{k(t+1)} \\ &\leq \sum_{t=1}^n ec(e^2c)^t (t/n)^{t+2} \leq \frac{ec}{n^2} \sum_{t=1}^{\infty} t^2 (e^2c)^t. \end{aligned}$$

The final series is finite because $e^2c < 1$. □

Wilson's Corollary 4 [18] applies to independent uniform proper k -clauses sampled with replacement. At the fixed probability levels $3/4$ and $1/4$, its assertion for every fixed $k \geq 3$ gives constants $d_k > 0$ and n_k such that, for $n \geq n_k$,

$$P_n(m_1) \geq \frac{3}{4}, \quad P_n(m_2) \leq \frac{1}{4} \implies m_2 - m_1 \geq d_k \sqrt{n} \quad (6.2)$$

The constants may depend on k . We spell out the conversion from this quantile-width statement to a variance bound, including possible atoms.

Proposition 6.2 (Linear variance lower bound). *Let L_n be positive integer caps of order $O(n)$ and suppose $P_n(L_n - 1) \leq 1/4$ for all sufficiently large n . Then*

$$\text{Var}(\min\{H_n, L_n\}) \geq c_k n$$

for all sufficiently large n , with $c_k > 0$. In particular, for every fixed $B > U_k$ and every fixed $k \geq 4$,

$$\text{Var}(H_n) = \Theta_k(n), \quad \text{Var}(T_{n,B}) = \Theta_{k,B}(n), \quad \text{Var}(\min\{H_n - 1, \lfloor Bn \rfloor\}) = \Theta_{k,B}(n). \quad (6.3)$$

For $k = 3$, the corresponding variances lie between positive constant multiples of n and $n \log n$.

Proof. Write $X = \min\{H_n, L_n\}$ and $G(j) = \mathbb{P}(X \leq j)$. Define the integer quantiles

$$a = \min\{j : G(j) \geq 1/4\}, \quad b = \min\{j : G(j) \geq 3/4\}.$$

The cap hypothesis gives $b \leq L_n - 1$. Thus $a - 1$ and b are below the cap and, by minimality,

$$P_n(a - 1) = 1 - G(a - 1) > 3/4, \quad P_n(b) = 1 - G(b) \leq 1/4.$$

Applying (6.2) to the endpoint probabilities above gives $b - a \geq d_k \sqrt{n} - 1$. Minimality of the quantiles also gives

$$\mathbb{P}(X \leq a) \geq \frac{1}{4}, \quad \mathbb{P}(X \geq b) = 1 - G(b - 1) > \frac{1}{4}.$$

For an independent copy X' , each of the disjoint events $\{X \leq a, X' \geq b\}$ and $\{X' \leq a, X \geq b\}$ has probability at least $1/16$ once $b > a$. Therefore

$$\text{Var}(X) = \frac{1}{2} \mathbb{E}(X - X')^2 \geq \frac{(b - a)^2}{16} \geq \frac{d_k^2}{64} n$$

for sufficiently large n . If $L_n = \lfloor Bn \rfloor$ and $B > U_k$, then

$$P_n(L_n - 1) \leq q_k^{-2} \rho_{k,B}^n = o(1)$$

by (5.1). The capped upper bounds proved above give the matching upper bounds. The exact identity

$$\min\{H_n - 1, L_n\} = \min\{H_n, L_n + 1\} - 1$$

proves the lower bound for last satisfiability, since its required condition is $P_n(L_n) \leq 1/4$. Finally $\text{Var}(H_n) \geq \text{Var}(\min\{H_n, L_n\})$ by Proposition 5.1; Corollary 5.2 supplies the uncapped upper bound. This completes the variance assertions of Theorem 1.1. $\square$

7 Three-clause cap and Poisson refinements

At clause size three, the cap identities connect the first-unsatisfiable statistic used in the variance proof to the last-satisfiable statistics used in the threshold companion [16]. We give their exact constants, then transfer concentration to the companion's auxiliary Poisson clause law, which allows repeated variables within a clause. The Poisson density interval and the cap on the hitting time play different roles and can be chosen independently. For $k = 3$, define the three distinct statistics

$$T = \min\{H_n, 10n\}, \quad W = \min\{H_n - 1, 10n\}, \quad V = \min\{H_n - 1, 16n\}.$$

With $\rho = 2(7/8)^{10} < 1$, Proposition 5.1 gives

$$W = T - 1 + \mathbf{1}_{\{F_{n,10n} \in \text{SAT}\}}, \quad (7.1)$$

$$V = T - 1 + R, \quad 0 \leq R \leq (6n + 1)\mathbf{1}_{\{F_{n,10n} \in \text{SAT}\}}. \quad (7.2)$$

Consequently $0 \leq \mathbb{E}V - (\mathbb{E}T - 1) \leq (6n + 1)\rho^n$ and $|\sqrt{\text{Var}(V)} - \sqrt{\text{Var}(T)}| \leq (6n + 1)\rho^{n/2}$. The sharper geometric bounds proved above are

$$\mathbb{E}R \leq 8\rho^n, \quad \mathbb{E}R^2 \leq 120\rho^n, \quad 0 \leq \text{Var}(V) - \text{Var}(T) \leq (160n + 120)\rho^n.$$

The pathwise inequality $W \leq 6n + 4n\mathbf{1}_{\{H_n > 6n\}}$ and the density-six first moment, with $\sigma = 2(7/8)^6 < 1$, give

$$\frac{\mathbb{E}W}{n} \leq 6 + 4\sigma^n = 6 + o(1). \quad (7.3)$$

The survival-sum bound $\mathbb{E}H_n \leq 6n + 8\sigma^n$ proves the same $6 + o(1)$ upper bound for T and V . Their normalized expectations are therefore all at most 7 for sufficiently large n .

On the density interval $0 \leq c \leq 8$, the three-clause Poisson law gives the following explicit transfer constants. Let $s_n(c)$ be the satisfiability probability for $\text{Pois}(cn)$ clauses, where the three variable choices within each clause are independent uniform choices with replacement and the signs are independent and fair. Internal repetitions and tautologies are allowed in this auxiliary law. The distinct-variable fraction is exactly

$$\vartheta_n = \frac{(n)_3}{n^3} = 1 - \frac{3}{n} + \frac{2}{n^2}.$$

Poisson thinning gives independent proper and repeated-variable clause collections. The proper count J has distribution $\text{Pois}(cn\vartheta_n)$, while the other count has mean

$$cn(1 - \vartheta_n) = c(3 - 2/n) \leq 24 \quad (0 \leq c \leq 8).$$

Requiring no repeated-variable clauses for a lower bound, and deleting them for an upper bound, gives

$$e^{-24}\mathbb{E}P_n(J) \leq s_n(c) \leq \mathbb{E}P_n(J), \quad cn - 24 \leq \mathbb{E}J \leq cn, \quad \text{Var}(J) \leq 8n. \quad (7.4)$$

Set $\omega_n = \mathbb{E}W/n$. The variance upper bound implies, in particular, $\mathbb{P}(|W - n\omega_n| \geq n^{11/12}) = O(n^{-1/6})$. Chebyshev's inequality for J and (7.4) therefore yield, uniformly for $c \in [0, 8]$,

$$c \leq \omega_n - 2n^{-1/12} \implies s_n(c) \geq e^{-24}/2, \quad (7.5)$$

$$c \geq \omega_n + 2n^{-1/12} \implies s_n(c) = O(n^{-1/6}). \quad (7.6)$$

Indeed, in the lower case $\mathbb{E}J \leq n\omega_n - 2n^{11/12}$, so $J \leq n\omega_n - n^{11/12}$ except with probability $O(n^{-5/6})$. In the upper case $\mathbb{E}J \geq n\omega_n + 2n^{11/12} - 24$, and the opposite deviation has the same bound. Apply concentration at the indicated integer counts; for counts beyond $10n$, use monotonicity at $\lceil n\omega_n + n^{11/12} \rceil < 10n$, valid by (7.3). These statements use the last-SAT cap $10n$ and Poisson density ceiling 8, giving the lower constant $e^{-24}/2$. The general transfer lemma of the threshold companion permits a cap density λ and a separate Poisson density ceiling D , each above a fixed upper bound for the normalized centers [16, Lemma 5.3]. Specializing it to $k = 3$ and $\lambda = D = 16$ uses the last-SAT statistic V , center $\mathbb{E}V/n$, and interval $c \in [0, 16]$. In that specialization the repetition bound is 48 and the lower constant is $e^{-48}/2$.

A Sharpness for arbitrary assignment sets

For each fixed integer $k \geq 3$, two exponents in Section 2 are optimal for estimates uniform over arbitrary assignment sets: the power $k/(k-1)$ of the inverse killing probability in the lifetime bound, and the power $k-1$ of the inverse block length in the one-clause replacement error. Both examples below use subcubes with many frozen coordinates. A common observation identifies when a block of clauses can kill such a subcube.

Write $r = k - 1$. We retain the proper clause law: a proper s -clause on u coordinates chooses s distinct coordinates uniformly and gives them independent fair signs. Different clauses are independent, with repetition of whole clauses allowed. Recall that $q_1(S)$ is the probability that one proper r -clause kills S , $R_t(S)$ is the probability that t proper k -clauses kill S , and $\tau_k(S) = \mathbb{E}H_k(S)$. For nonempty S , $H_k(S)$ is the least positive index t for which the first t proper k -clauses jointly kill S ; the convention for the empty set is $H_k(\emptyset) = 0$.

For integers $u \geq b \geq k$, consider the nonempty subcube

$$S_{u,b} = \{x \in \{0, 1\}^u : x_1 = \cdots = x_b = 0\}.$$

Its first b coordinates are frozen and the others are free. If $h_s(S_{u,b})$ denotes the probability that one proper s -clause kills this subcube, the frozen-coordinate formula (2.2) reads

$$h_s(S_{u,b}) = \frac{\binom{b}{s}}{2^s(u)_s} \quad (1 \leq s \leq u), \quad q_1(S_{u,b}) = h_r(S_{u,b}), \quad (\text{A.1})$$

where $(a)_s = a(a-1)\cdots(a-s+1)$. In particular,

$$\frac{h_k(S_{u,b})}{q_1(S_{u,b})} = \frac{b-r}{2(u-r)}. \quad (\text{A.2})$$

For every integer $t \geq 1$, the same subcube satisfies

$$1 - (1 - h_k(S_{u,b}))^t \leq R_t(S_{u,b}) \leq t h_k(S_{u,b}) + \binom{t}{2} \frac{k^2}{u}. \quad (\text{A.3})$$

The lower bound records the independent chances that a single clause kills the whole subcube. To prove the upper bound, restrict all t clauses to the frozen values. Each restricted clause is already true, is the empty false clause, or is a nonempty clause on free coordinates. If no restricted clause is empty and false, and no two original clauses share a free coordinate, then the remaining nonempty clauses use disjoint sets of variables. Choose one literal in each and set it true, then assign the other free coordinates arbitrarily. The resulting assignment lies in $S_{u,b}$ and satisfies the block. Thus a block can kill the subcube only if an individually killing clause occurs or a pair of clauses shares a free coordinate. For a fixed free coordinate, the probability that both independent proper clauses contain it is $(k/u)^2$. Summing over the $u-b$ free coordinates bounds the probability for a given pair by k^2/u . A union bound over clauses and pairs proves the upper bound in (A.3).

Proposition A.1 (Optimality of the lifetime exponent). *Fix an integer $k \geq 3$, and put $\beta = k/(k-1)$. For every real $\gamma < \beta$ and all constants $D, A > 0$, there exist arbitrarily large integers $u \geq k$ and nonempty sets $S \subseteq \{0,1\}^u$ such that*

$$q_1(S) \geq A u^{-(k-1)}, \quad \tau_k(S) > D q_1(S)^{-\gamma}.$$

Consequently, the killing-probability exponent β in Proposition 2.3 cannot be decreased in a bound uniform over arbitrary assignment sets, even above any fixed multiple of the scale $u^{-(k-1)}$.

Proof. Let the integer $N \geq 2$ tend to infinity, and put

$$u = N^{2k+1}, \quad b = N^{2k}, \quad S = S_{u,b}.$$

Since k is fixed and $b/u = N^{-1}$, (A.1) gives

$$a_N := q_1(S) \sim 2^{-r} N^{-r}, \quad h_N := h_k(S) \sim 2^{-k} N^{-k}. \quad (\text{A.4})$$

Choose $t_N = \lfloor 1/(4h_N) \rfloor$. This is positive for all sufficiently large N , and $t_N = \Theta_k(N^k)$. A set survives the first t_N clauses exactly when $H_k(S) > t_N$, so (A.3) yields

$$\mathbb{P}(H_k(S) > t_N) = 1 - R_{t_N}(S) \geq 1 - t_N h_N - \binom{t_N}{2} \frac{k^2}{u} \geq \frac{3}{4} - o(1). \quad (\text{A.5})$$

Here $t_N h_N \leq 1/4$, while $t_N^2/u = O_k(N^{-1})$. It follows that

$$\tau_k(S) \geq t_N \mathbb{P}(H_k(S) > t_N) = \Omega_k(N^k).$$

For the reverse bound, let J be the index of the first clause that kills the original set S by itself. The cumulative sequence kills S no later than J . Since J has the geometric distribution with success probability $h_N > 0$,

$$\tau_k(S) \leq \mathbb{E}J = h_N^{-1} = O_k(N^k).$$

Thus

$$\tau_k(S) = \Theta_k(N^k) = \Theta_k(a_N^{-\beta}).$$

For each fixed $\gamma < \beta$, the ratio $\tau_k(S)/a_N^{-\gamma}$ tends to infinity, since $k - r\gamma > 0$. Finally,

$$a_N u^r \sim 2^{-r} N^{2kr} = 2^{-r} b^r \longrightarrow \infty.$$

For all sufficiently large N , the two displayed requirements therefore hold for the prescribed D and A . The values of u are arbitrarily large. $\square$

Proposition A.2 (Optimality of the block-error exponent). *Fix an integer $k \geq 3$. For integers $g \geq 2$, there are nonempty sets $S_g \subseteq \{0, 1\}^{u_g}$, with $u_g \geq k$, for which*

$$q_1(S_g) - R_g(S_g) \sim 2^{-k} g^{-(k-1)} \quad (g \rightarrow \infty).$$

In particular, no error $o(g^{-(k-1)})$, uniform over u and the assignment set, can replace the error in Lemma 2.1.

Proof. For every integer $g \geq 2$, put

$$u_g = g^{k+2}, \quad b_g = g^{k+1}, \quad S_g = S_{u_g, b_g},$$

and abbreviate $a_g = q_1(S_g)$ and $h_g = h_k(S_g)$. Equations (A.1) and (A.2) give

$$a_g \sim 2^{-r} g^{-r}, \quad \frac{gh_g}{a_g} = \frac{g(b_g - r)}{2(u_g - r)} \longrightarrow \frac{1}{2}.$$

The upper bound in (A.3), and the two-term inclusion–exclusion lower bound for the independent individual killing events, give

$$gh_g - \binom{g}{2} h_g^2 \leq R_g(S_g) \leq gh_g + \binom{g}{2} \frac{k^2}{u_g}.$$

The collision error is $O_k(g^{-k}) = o(a_g)$. Also $h_g = \Theta_k(g^{-k})$, so the inclusion–exclusion error is $O_k(g^{-2(k-1)}) = o(a_g)$. Hence

$$R_g(S_g) = gh_g + o(a_g), \quad q_1(S_g) - R_g(S_g) = \left(\frac{1}{2} + o(1)\right)a_g \sim 2^{-k} g^{-r}.$$

This is the stated asymptotic. $\square$

These constructions delimit estimates uniform over arbitrary assignment sets. They leave open stronger lifetime estimates for classes of sets with additional structure, and variance arguments using information beyond the killing probability. In particular, they do not show that the logarithm in the three-clause variance upper bound is necessary.

B Comparing finite-size centers with proper clauses

The variance estimate controls fluctuations at one size. We now compare the corresponding centers at different sizes, using proper clauses at every step. The comparison keeps the weights of the two variable blocks; this is what makes its error uniform even when one block is much larger than the other. Once these probability comparisons are established, the deterministic balanced-comparison lemma of the threshold companion turns them into convergence. This gives a second route to convergence from a concentration estimate, without using the companion's limiting-threshold theorem as Section 5 does. The theorem below takes concentration as an input, so it also applies to the independent, weaker variance proof in Appendix C.

Interpolation between a system and independent parts appears in the thermodynamic-limit work of Guerra and Toninelli [13] and the diluted-system work of Franz and Leone [10] and Franz, Leone, and Toninelli [11]. Those works concern free energies under their respective model hypotheses; they provide methodological context for the size comparison. The one-clause interpolation here uses frozen variables, as in the satisfiability interpolation of Bayati, Gamarnik, and Tetali [3, Section 4, Proposition 2]. Their clause law samples variable tuples with replacement. Here the clauses have distinct variables, so the exact killing probabilities contain falling factorials. We control the resulting error locally before summing it over the trials.

B.1 A center theorem with an explicit concentration input

For fixed $k \geq 3$ and $n \geq k$, let H_n be the index of the first unsatisfiable prefix of independent uniform proper k -clauses on n variables, and write $P_n(j) = \mathbb{P}(H_n > j)$ for integers $j \geq 0$. Fix a real $B > 0$ and set

$$L_n = \lfloor Bn \rfloor, \quad T_n = \min\{H_n, L_n\}, \quad \mu_n = \mathbb{E}T_n, \quad e_n = \mu_n/n.$$

The endpoint for this first-unsatisfiable convention is strict:

$$P_n(j) = \mathbb{P}(T_n > j) \quad (0 \leq j < L_n). \quad (\text{B.1})$$

At $j = L_n$, the probability on the right is zero, whereas $P_n(L_n)$ need not vanish. All statements below use only indices strictly below the cap when applying (B.1).

Theorem B.1 (Proper-clause center comparison from concentration). *Fix $k \geq 3$, $B > 0$, and $0 < \delta < 1/2$. Suppose that there are constants $0 < a \leq A < B$ and numbers $\eta_n \geq 0$ tending to zero such that, for all sufficiently large n ,*

$$a \leq e_n \leq A, \quad \mathbb{P}(|T_n - \mu_n| \geq n^{1-\delta}) \leq \eta_n. \quad (\text{B.2})$$

Then there is a finite constant $K \geq 0$ such that, for all sufficiently large $n = \min\{n_1, n_2\}$, with no restriction on n_1/n_2 ,

$$e_{n_1+n_2} \geq \min\{e_{n_1}, e_{n_2}\} - Kn^{-\delta}, \quad (\text{B.3})$$

and, for all sufficiently large n ,

$$e_{n+1} \geq e_n - Kn^{-\delta}. \quad (\text{B.4})$$

Moreover e_n converges to a number $\alpha \in [a, A]$, and

$$P_n(\lfloor cn \rfloor) \longrightarrow \begin{cases} 1, & 0 \leq c < \alpha, \\ 0, & c > \alpha. \end{cases} \quad (\text{B.5})$$

For every sufficiently large s and every $N \geq s^2$, one has the one-sided estimate

$$e_N \geq e_s - K(1 + B_\delta)s^{-\delta}, \quad B_\delta = \frac{(3/2)^\delta}{1 - (2/3)^\delta}. \quad (\text{B.6})$$

The theorem allows an arbitrary vanishing concentration error; it does not require a specified polynomial rate. In particular, Chebyshev's inequality supplies (B.2) whenever

$$\text{Var}(T_n) = o(n^{2-2\delta}). \quad (\text{B.7})$$

The proof uses concentration twice: to make the two local formulas likely to be satisfiable, and to turn a likely satisfiable diluted formula back into a lower bound for the center at the combined size.

B.2 Local clauses and diluted global clauses

Lemma B.2 (Proper-clause trial comparison). *Partition $N = n_1 + n_2$ variables into blocks of sizes $n_1, n_2 \geq k$, and put $w_j = n_j/N$ and $n = \min\{n_1, n_2\}$. Fix an integer $M \geq 0$ and $0 < \varepsilon < 1$. Each of M independent local trials chooses block j with probability w_j and inserts an independent uniform proper k -clause in that block. Each of M independent diluted global trials inserts an independent uniform proper k -clause on all N variables with probability $1 - \varepsilon$, and inserts no clause otherwise. Let Q_{loc} and Q_{dil} be the respective probabilities that the resulting formulas are satisfiable. Then*

$$Q_{\text{dil}} \geq Q_{\text{loc}} - C_k M \varepsilon^{-k} \sum_{j=1}^2 \frac{w_j}{n_j^k}, \quad C_k = \frac{[k(k-1)]^k}{2^k}, \quad (\text{B.8})$$

$$Q_{\text{dil}} \geq Q_{\text{loc}} - \frac{C_k M}{\varepsilon^k n^k}. \quad (\text{B.9})$$

For $N = n + 1$, if all local trials instead use a fixed main set of $n \geq k$ variables, then

$$Q_{\text{dil}} \geq Q_{\text{loc}} - \frac{D_k M}{\varepsilon^k n^k}, \quad D_k = \frac{k^{2k}}{2^k}. \quad (\text{B.10})$$

These statements impose no restriction on a block-size ratio.

Proof. Condition on every trial except the one being replaced. If the residual formula is unsatisfiable, both conditional satisfiability probabilities are zero. Otherwise let S be its nonempty solution set, and call a variable frozen when it has the same value in every member of S . Among any specified $v \geq k$ variables of which b are frozen, a proper k -clause kills S with probability

$$h_k(b, v) = \frac{(b)_k}{2^k (v)_k}.$$

Indeed, each of the selected variables must be frozen and each sign must oppose its frozen value. With $x = b/v$ and $y_+ = \max\{y, 0\}$,

$$\frac{(x - (k-1)/v)_+^k}{2^k} \leq h_k(b, v) \leq \frac{x^k}{2^k}. \quad (\text{B.11})$$

For $b \geq k$, each factor $(b-r)/(v-r)$, $0 \leq r < k$, is at most b/v and at least $(b-k+1)/v$. For $b < k$, both the middle expression and the positive part on the left vanish. This proves the bounds.

Let β_j be the frozen fraction in block j , always with respect to the entire residual set S ; its defining formula may contain clauses crossing the two blocks. By convexity and (B.11), the killing probability of a nondiluted global clause is at most

$$2^{-k} \left(\sum_j w_j \beta_j \right)^k \leq 2^{-k} \sum_j w_j \beta_j^k,$$

whereas the local killing probability is at least $2^{-k} \sum_j w_j (\beta_j - (k-1)/n_j)_+^k$. If $\beta_j \geq k(k-1)/(\varepsilon n_j)$, Bernoulli's inequality gives

$$(\beta_j - (k-1)/n_j)_+^k \geq (1 - \varepsilon/k)^k \beta_j^k \geq (1 - \varepsilon) \beta_j^k.$$

Below that threshold, $\beta_j^k \leq [k(k-1)]^k / (\varepsilon^k n_j^k)$. It follows that the diluted global killing probability exceeds the local one by at most $C_k \varepsilon^{-k} \sum_j w_j / n_j^k$. The estimate holds for every residual formula. Replacing the trials one at a time, conditioning on all other independent trials at each step, and then averaging and telescoping proves (B.8). Since $n_j \geq n$ and $w_1 + w_2 = 1$, it also proves the coarser bound.

For neighboring sizes, let b count the frozen variables in the main n -set and let $x \in \{0, 1\}$ indicate whether the extra variable is frozen. The global frozen fraction $\gamma = (b+x)/(n+1)$ obeys $b/n \geq \gamma - 1/(n+1)$. The local killing probability is therefore at least $2^{-k} (\gamma - k/n)_+^k$, while the nondiluted global one is at most $2^{-k} \gamma^k$. Splitting at $\gamma = k^2/(\varepsilon n)$ and applying the same Bernoulli inequality bounds the per-trial loss by $D_k/(\varepsilon^k n^k)$. Telescoping gives (B.10). $\square$

At $k = 3$ the constants are exactly $C_3 = 27$ and $D_3 = 729/8$. The stronger weighted bound will be used for unequal blocks. The coarser bound remains useful when the block sizes are known to be comparable.

B.3 From a random clause count to a center

We first record the integer step needed in both center comparisons. From (B.2) and (B.1), for every integer $0 \leq j < L_n$ and all sufficiently large n ,

$$\begin{aligned} j \leq \mu_n - n^{1-\delta} &\implies P_n(j) \geq 1 - \eta_n, \\ j \geq \mu_n + n^{1-\delta} &\implies P_n(j) \leq \eta_n. \end{aligned} \tag{B.12}$$

Because $\eta_n \rightarrow 0$, its tail supremum $\bar{\eta}_n = \sup_{v \geq n} \eta_v$ also tends to zero. This will provide uniformity when one block is arbitrarily larger than the other.

Suppose, for a fixed $a_0 > 0$, that $a_0 N \leq M \leq AN$, $0 < \varepsilon \leq 1/2$, and $Q_{\text{dil}} \geq 1 - \zeta$. The number of inserted clauses is $J \sim \text{Bin}(M, 1 - \varepsilon)$. Conditional on J , the inserted clauses have exactly the independent proper-clause law, so $Q_{\text{dil}} = \mathbb{E} P_N(J)$. Put

$$j = \lfloor (1 - \varepsilon)M - N^{1-\delta} \rfloor.$$

For all sufficiently large N , the lower bound on M gives $j \geq 0$, and $j \leq M \leq AN < L_N$ because $A < B$. Also $\text{Var}(J) \leq AN$, whence

$$\mathbb{P}(J < j) \leq AN^{-1+2\delta}, \quad Q_{\text{dil}} \leq P_N(j) + \mathbb{P}(J < j).$$

The second inequality uses monotonicity of P_N . If $\zeta + AN^{-1+2\delta} + \eta_N < 1$, the upper implication in (B.12) forces $j < \mu_N + N^{1-\delta}$. The floor in the definition of j then gives

$$e_N > \frac{(1-\varepsilon)M}{N} - 2N^{-\delta} - N^{-1} \geq \frac{(1-\varepsilon)M}{N} - 3N^{-\delta}. \quad (\text{B.13})$$

We will apply this implication with all three errors tending to zero.

Proof of Theorem B.1. For the two-block comparison, put $N = n_1 + n_2$, $n = \min\{n_1, n_2\}$, $e_* = \min\{e_{n_1}, e_{n_2}\}$, and choose

$$\varepsilon = n^{-\delta}, \quad M = \lfloor (e_* - 2\varepsilon)N \rfloor.$$

For sufficiently large n , the anchors imply $(a/4)N \leq M \leq AN$ and $\varepsilon \leq 1/2$. In the local model let X_j count the clauses in block j . Its marginal law is binomial, and

$$\mathbb{E}X_j = Mn_j/N \leq \mu_{n_j} - 2\varepsilon n_j \leq \mu_{n_j} - 2n_j^{1-\delta}, \quad \text{Var}(X_j) \leq An_j.$$

Thus $\mathbb{P}(X_j > \mu_{n_j} - n_j^{1-\delta}) \leq An_j^{-1+2\delta}$. Conditional on the block choices, the clauses inside the two blocks have independent proper laws. On the event $X_j \leq \mu_{n_j} - n_j^{1-\delta}$, the integer count lies below L_{n_j} for large n_j , because $\mu_{n_j} \leq An_j$ and $A < B$. The lower implication in (B.12) and a union bound therefore give

$$Q_{\text{loc}} \geq 1 - \sum_{j=1}^2 (An_j^{-1+2\delta} + \eta_{n_j}).$$

The right-hand error is at most $2An^{-1+2\delta} + 2\bar{\eta}_n$, which tends to zero uniformly over all pairs of sizes with minimum n .

The weighted replacement loss has the same uniformity:

$$\begin{aligned} C_k M \varepsilon^{-k} \sum_j \frac{w_j}{n_j^k} &= C_k \frac{M}{N} \varepsilon^{-k} \sum_j n_j^{1-k} \\ &\leq 2AC_k n^{1-k+k\delta} = o(1). \end{aligned} \quad (\text{B.14})$$

Here $1 - k + k\delta < 0$ for $k \geq 3$ and $\delta < 1/2$. The factor N has canceled, so no upper bound on N/n is used. Lemma B.2 therefore gives $Q_{\text{dil}} \geq 1 - \zeta_n$, where $\zeta_n \rightarrow 0$ uniformly over these pairs. Since $N \geq 2n$, the count error at size N and $\eta_N \leq \bar{\eta}_n$ also vanish uniformly. Applying (B.13), and keeping the floor in M , yields

$$e_N \geq (1-\varepsilon)(e_* - 2\varepsilon - N^{-1}) - 3N^{-\delta} \geq e_* - Kn^{-\delta},$$

where the last inequality uses $e_* \leq A$, $N \geq n$, and $\delta < 1$. This proves (B.3) for arbitrary size ratios.

For the neighboring comparison take $N = n+1$, put $\varepsilon = n^{-\delta}$, and let $M = \lfloor \mu_n - 2n^{1-\delta} \rfloor$ local trials all use the main n variables. Again $(a/4)N \leq M \leq AN$ for sufficiently large n . The local formula has exactly the law at size n and count M , so $Q_{\text{loc}} \geq 1 - \eta_n$ by (B.12). The neighboring replacement loss is

$$\frac{D_k M}{\varepsilon^k n^k} = O_{k,A}(n^{1-k+k\delta}) = o(1).$$

Equation (B.13) now gives

$$e_{n+1} \geq (1 - \varepsilon) \frac{ne_n - 2n^{1-\delta} - 1}{n+1} - 3(n+1)^{-\delta} \geq e_n - Kn^{-\delta}.$$

For the last step, use $e_n \leq A$ and $1 - (1 - \varepsilon)n/(n+1) \leq \varepsilon + 1/(n+1)$. This proves (B.4).

It remains to explain the deterministic step and check its inputs. The geometric summation of errors parallels Abbe and Montanari's approximate sum-superadditivity argument [1, arXiv version, Lemma 8 and Remark 3]. The required minimum comparison is the threshold companion's Lemma 5.4 ("Balanced comparisons") [16]. For an integer $n_0 \geq 1$, it states that if a bounded real sequence $(u_n)_{n \geq n_0}$ satisfies

$$u_{r+s} \geq \min\{u_r, u_s\} - K \min(r, s)^{-\delta} \quad \text{for } 1/3 \leq r/s \leq 3, \quad u_{s+1} \geq u_s - Ks^{-\delta},$$

for all $r, s \geq n_0$ in the indicated ratio range and all $s \geq n_0$, respectively, with fixed $K \geq 0$ and $\delta > 0$, then it converges and, for every $s \geq \max\{n_0, 2\}$ and every $N \geq s^2$, $u_N \geq u_s - K(1 + B_\delta)s^{-\delta}$ with B_δ as above. The lemma is a deterministic minimum comparison: it uses leaves of sizes $s, s+1$, and its binary tree has child-size ratios at most three. Choose n_0 beyond the cutoffs in our anchors and two comparisons. Then the boundedness in (B.2), the comparable-size part of (B.3), and (B.4) verify every hypothesis with $u_n = e_n$. The lemma gives (B.6) and convergence to some $\alpha \in [a, A]$.

For $0 \leq c < \alpha$, eventually $\lfloor cn \rfloor \leq \mu_n - n^{1-\delta}$ and $\lfloor cn \rfloor < L_n$. For $\alpha < c < B$, eventually $\mu_n + n^{1-\delta} \leq \lfloor cn \rfloor < L_n$. In both cases the fixed positive density gap eventually exceeds $n^{1-\delta} + 1$, accounting for the floor. Apply (B.12). Since $\alpha \leq A < B$, choose one fixed $d \in (A, B)$; monotonicity extends the zero limit from d to all $c \geq B$. This proves (B.5). $\square$

The comparison (B.6) is one-sided. It proves convergence of the finite-size means, but gives no two-sided estimate for $\mu_n - \alpha n$ on the fluctuation scale of the variance theorem.

B.4 Checking the inputs for the variance estimates

For the main application put $B_k = 2^{k+1}$ and take $B = B_k$; at $k = 3$ we may also take $B = 10$. We verify the anchors without assuming threshold convergence. Let $q_k = 1 - 2^{-k}$. The assignment first moment gives $P_n(j) \leq 2^n q_k^j$, hence, for every positive integer d ,

$$\mathbb{E}H_n = \sum_{j=0}^{\infty} P_n(j) \leq dn + 2^k (2q_k^d)^n. \quad (\text{B.15})$$

For $d = 2^k$, one has $2q_k^{2^k} < 2/e < 1$. For $k = 3$, the sharper choice $d = 6$ satisfies $2(7/8)^6 < 1$. Since $T_n \leq H_n$, it follows that eventually

$$e_n \leq A_k < B, \quad A_k = \begin{cases} 7, & k = 3, \\ 2^k + 1, & k \geq 4. \end{cases} \quad (\text{B.16})$$

For the positive anchor, fix $0 < c < \min\{e^{-2}, B\}$ and let $j = \lfloor cn \rfloor$. Lemma 6.1 gives $P_n(j) = 1 - O_c(n^{-2})$ for the same independent proper-clause model. Since $c < B$, one has $j < L_n$ for sufficiently large n . Therefore $\mu_n \geq jP_n(j) = cn - o(n)$, and with $c_0 = c/2$ we obtain the full anchors

$$c_0 \leq e_n \leq A_k < B \quad \text{for all sufficiently large } n. \quad (\text{B.17})$$

The capped variance theorem gives

$$\frac{\text{Var}(T_n)}{n^{2-2\delta}} = \begin{cases} O(n^{-1+2\delta} \log n), & k = 3, \\ O_k(n^{-1+2\delta}), & k \geq 4, \end{cases}$$

which tends to zero for every fixed $0 < \delta < 1/2$. Theorem B.1 now applies with $a = c_0$ and $A = A_k$. Its strict-side limit is the same α_k as in the threshold companion [16, Theorem 1.1]: if the two limits differed, a density strictly between them would be assigned opposite limiting probabilities. In particular, the proper-clause comparisons (B.3)–(B.4) hold for every such fixed δ .

The weaker polynomial variance estimate of the threshold companion [16, Proposition 3.3] can be used as a separate input. For $a = B_k n$ and the same last-SAT cap $b = a$, the exact identity in Proposition 5.1 is

$$\min\{H_n - 1, a\} = \min\{H_n, a\} - 1 + \mathbf{1}_{\{H_n > a\}}.$$

That proposition gives $\text{Var}(\min\{H_n, a\}) \leq \text{Var}(\min\{H_n - 1, a\})$. Consequently the companion's $O_k(n^{1+2/k})$ bound verifies (B.7) for every $0 < \delta < (k-2)/(2k)$, including the companion's stated choice $\delta = (k-2)/(4k)$. This substitution uses the companion's own variance estimate. The range $\delta < 1/2$ for the local variance bounds is also strict; no endpoint assertion is made.

C A three-clause proof using prefix exposure

We give a second variance estimate for three-clause formulas, revealing clause types only through the current prefix. The argument truncates the number of clauses testing each variable of an omitted clause at $\lceil \log n \rceil$, and pays separately for clauses meeting two or more of those variables. It gives an $O(n \log^4 n)$ bound. The cube of the incidence cutoff contributes three logarithmic factors; the critical occupation integral contributes the fourth. The main argument in Sections 3–4 instead averages the bounded third moment of the actual incidence count, giving $O(n \log n)$. Comparing the two proofs shows where that improvement enters. The replacement step below also gives a distinct finite-variable estimate: replacing one two-clause on u variables by g three-clauses loses at most $64(g^{-2} + u^{-2})$ in killing probability.

Throughout this section, $C_1, C_2, \dots$ are independent uniform proper three-clauses on $n \geq 3$ variables: a clause uses three distinct variables and independent fair signs, while whole clauses may repeat. Write H_n for the index of the first unsatisfiable prefix and put

$$L = 10n, \quad T_n = \min\{H_n, L\}, \quad m_n = \mathbb{E}T_n.$$

Theorem C.1 (Variance from prefix exposure). *There is an absolute constant C such that, for all integers $n \geq 3$,*

$$\text{Var}(T_n) \leq Cn(\log n)^4.$$

Consequently, for every $t > 0$,

$$\mathbb{P}(|T_n - m_n| \geq t\sqrt{n}(\log n)^2) \leq Ct^{-2}.$$

The proof has two uses of the same lifetime estimate. It bounds the remaining duration of a deletion event and the total time spent at a given level of the probability of killing the surviving assignments. Three independent root-flip tests connect those two bounds.

C.1 Replacing pairs by triples on a finite variable set

For an integer $u \geq 3$ and $S \subseteq \{0, 1\}^u$, a collection of clauses *kills* S if no member of S satisfies all of them. A uniform proper r -clause here is chosen from the $2^r \binom{u}{r}$ signed clauses on r distinct variables. For an integer $d \geq 1$, let $q_d(S)$ be the probability that d independent uniform proper two-clauses kill S . Let $\tau(S)$ be the expected length of the first prefix of independent uniform proper three-clauses whose conjunction kills S , and set $\tau(\emptyset) = 0$. For nonempty S , this expectation is finite because the probability of survival after m clauses is at most $|S|(7/8)^m$.

Replacing a shorter constraint by native clauses has a qualitative predecessor in Friedgut's half-cube argument [12, Lemma 5.7]. The residual-set conditioning used below is the averaged replacement method of Carenini [5, Theorem 4.3, Lemma 6.1, and Corollary 6.2]; the following bound supplies the finite-variable pair-to-triple estimate needed here.

Lemma C.2 (One-pair replacement). *For every integer $u \geq 3$, every $S \subseteq \{0, 1\}^u$, and every integer $g \geq 1$, if the g triples are independent uniform proper three-clauses on these u variables, then*

$$\mathbb{P}(\text{the } g \text{ triples kill } S) \geq q_1(S) - 64(g^{-2} + u^{-2}). \quad (\text{C.1})$$

Proof. If $S = \emptyset$, both killing probabilities are one. Suppose that $S \neq \emptyset$, and let b be the number of variables taking a common value on all members of S . A single clause kills S exactly when every one of its variables is among these b frozen variables and every sign opposes the common value. Thus the killing probability of a proper r -clause is

$$h_r = \frac{(b)_r}{2^r \binom{u}{r}}, \quad r = 2, 3,$$

where $(x)_r = x(x-1)\cdots(x-r+1)$, and $(b)_r = 0$ for integers $0 \leq b < r$. For $b \geq 3$,

$$\frac{h_3}{h_2^{3/2}} = \frac{b-2}{\sqrt{b(b-1)}} \frac{\sqrt{u(u-1)}}{u-2} \geq \frac{1}{4}.$$

Indeed, the first factor is at least $(b-2)/b \geq 1/3$, and the second is at least one.

Put $h = h_2 \leq 1/4$. The block kills S whenever one of its clauses kills S by itself, so its killing probability is at least $1 - (1 - h_3)^g \geq 1 - e^{-gh_3}$. If $b \geq 3$ and $(g/4)\sqrt{h} \geq 2$, then $gh_3 \geq 2h$, and this probability is at least $1 - e^{-2h} \geq h$. The last inequality follows because $1 - e^{-2h} - h$ vanishes at zero and has nonnegative derivative on $[0, 1/4]$. In the remaining case with $b \geq 3$, $h \leq 64g^{-2}$. If $b < 3$, then

$$h \leq \frac{1}{2u(u-1)} \leq u^{-2}.$$

The nonnegativity of the block killing probability now proves (C.1) in every case. $\square$

Remark C.3. For $k = 3$, the cutoff-free estimate in Lemma 2.1 has error $256g^{-2}$. Neither that error nor $64(g^{-2} + u^{-2})$ is uniformly smaller: the latter is smaller for $g < \sqrt{3}u$, the former is smaller for $g > \sqrt{3}u$, and they are equal when $g = \sqrt{3}u$. The two estimates therefore retain different quantitative information.

Proposition C.4 (Lifetime above the finite-variable cutoff). *Let $K = 64$. There is an absolute constant K_1 such that for every integer $u \geq 3$, every nonempty $S \subseteq \{0, 1\}^u$, every integer $d \geq 1$, and every a with $4Kd/u^2 \leq a \leq 1$,*

$$q_d(S) \geq a \implies \tau(S) \leq K_1 d^{3/2} a^{-3/2}. \quad (\text{C.2})$$

Proof. Replace d independent two-clauses one at a time by independent blocks of g proper three-clauses. At each step condition on the clauses in every other position, including the blocks already inserted. Their surviving assignments form a set $S' \subseteq S$. The pair being removed and the new block are independent of this conditioning. Lemma C.2, including its empty-set case, bounds the conditional loss by $K(g^{-2} + u^{-2})$. Averaging and telescoping over the replacements gives

$$\mathbb{P}(dg \text{ independent uniform proper three-clauses kill } S) \geq q_d(S) - Kd(g^{-2} + u^{-2}). \quad (\text{C.3})$$

Choose $g = \lceil \sqrt{4Kd/a} \rceil$. Each of Kd/g^2 and Kd/u^2 is at most $a/4$, so a block of dg triples kills the original set S with probability at least $a/2$. In an infinite independent sequence, disjoint blocks have independent events of killing that original set. Their first successful block has expected index at most $2/a$, and the cumulative sequence kills S by the end of that block. Therefore

$$\tau(S) \leq \frac{2dg}{a} \leq 8\sqrt{K} d^{3/2} a^{-3/2},$$

where $g \leq 2\sqrt{4Kd/a}$ follows from $d \geq 1$ and $a \leq 1$. This proves the assertion, for example with $K_1 = 64$. $\square$

C.2 A deleted clause and an unexposed future

We now apply the lifetime estimate to the delay caused by omitting one clause. We first assume n is sufficiently large, in particular $n \geq 6$; the final constant will cover the remaining sizes. Keep the original clause indices, and for $1 \leq i \leq L$ and integer $m \geq 0$ put

$$B_m^{(i)} = \bigwedge_{\substack{1 \leq j \leq m \\ j \neq i}} C_j, \quad T^{(i)} = \min\{\inf\{m \in \mathbb{Z}_{\geq 0} : B_m^{(i)} \notin \text{SAT}\}, L\}, \quad D_i = T^{(i)} - T_n,$$

with $\inf \emptyset = \infty$. Then $D_i \geq 0$, and $T^{(i)}$ does not use coordinate C_i . The coordinate-omission inequality, Lemma 3.1, gives

$$\text{Var}(T_n) \leq \sum_{i=1}^L \mathbb{E} D_i^2. \quad (\text{C.4})$$

Its martingale proof applies to precisely these index-preserving deletions and to the cap $L = 10n$.

Fix i and condition on any particular clause C_i . Until the end of the variance proof, probabilities refer to this conditional law; the other clauses remain independent and uniform. We call indices other than i the baseline indices for this deletion. Let R be the three variables of C_i , called its *roots*, and let $t \in \{0, 1\}^R$ be their unique values that falsify C_i . Put $u = n - 3$, abbreviate $B_m = B_m^{(i)}$, and let $\mathcal{F}_m = \sigma(C_j : 1 \leq j \leq m, j \neq i)$ under the conditional law. In particular, $u \geq 3$.

For $i \leq m < L$, define

$$A_m = \{T_n \leq m < T^{(i)}\} = \{B_m \in \text{SAT}, B_m \wedge C_i \notin \text{SAT}\}. \quad (\text{C.5})$$

On A_m , every solution of B_m has root values t . The two processes agree before index i , so

$$D_i = \sum_{m=i}^{L-1} \mathbf{1}_{\{A_m\}}. \quad (\text{C.6})$$

In particular $D_L = 0$. For every $m \geq i$, let $S_m \subseteq \{0, 1\}^u$ be the nonroot restrictions of solutions of B_m with root values t . The sets are decreasing, $\mathcal{F}_m$ -measurable, and nonempty on A_m .

A clause is *ordinary* if it uses only nonroots. A fresh clause is ordinary with probability

$$p_0 = \frac{\binom{u}{3}}{\binom{n}{3}},$$

which is bounded below by a positive absolute constant for sufficiently large n . Let U_m be the number of trials after index m , including the last trial, until the ordinary clauses among them kill the original set S_m ; put $U_m = 0$ when $S_m = \emptyset$. These trials may continue past L . Conditional on $\mathcal{F}_m$, the future ordinary clauses are independent uniform proper triples on the nonroots. Their arrival gaps, counting the arrival trial, are independent geometric variables of mean $1/p_0$, independent of the ordinary clauses themselves. Conditioning first on the latter clauses and summing the means of the gaps through their killing time proves

$$\mathbb{E}[U_m \mid \mathcal{F}_m] = \frac{\tau(S_m)}{p_0}. \quad (\text{C.7})$$

The assertion uses the fact that only the prefix has been revealed; for $m \geq i$, all trials after m remain fresh even after fixing C_i .

Lemma C.5 (Remaining deletion time). *Set $\mathcal{W}_m = \min\{L, \tau(S_m)/p_0\}$. For $i \leq m < L$, on A_m ,*

$$\mathbb{E} \left[\sum_{\ell=m+1}^{L-1} \mathbf{1}_{\{A_\ell\}} \mid \mathcal{F}_m \right] \leq \mathcal{W}_m.$$

Consequently,

$$\mathbb{E} D_i^2 \leq 3 \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{\{A_m\}} \mathcal{W}_m]. \quad (\text{C.8})$$

Proof. On A_m , any later solution of B_ℓ still has root values t , and its nonroot part lies in S_m . Once the subsequent ordinary clauses kill S_m , no such solution remains. Thus A_ℓ is impossible for $\ell \geq m + U_m$. The number of later indices is at most both U_m and L ; its conditional expectation is therefore at most $\min\{\mathbb{E}[U_m \mid \mathcal{F}_m], L\}$. Equation (C.7) proves the first assertion.

On A_m , the nonempty set S_m requires at least one triple to be killed, so $\mathcal{W}_m \geq 1$. Expanding (C.6), conditioning each cross term on its earlier history, and using the first assertion gives

$$\begin{aligned} \mathbb{E} D_i^2 &= \sum_m \mathbb{E} \mathbf{1}_{\{A_m\}} + 2 \sum_m \mathbb{E} \left[\mathbf{1}_{\{A_m\}} \mathbb{E} \left[\sum_{\ell=m+1}^{L-1} \mathbf{1}_{\{A_\ell\}} \mid \mathcal{F}_m \right] \right] \\ &\leq \sum_m \mathbb{E}[\mathbf{1}_{\{A_m\}} (1 + 2\mathcal{W}_m)] \leq 3 \sum_m \mathbb{E}[\mathbf{1}_{\{A_m\}} \mathcal{W}_m], \end{aligned}$$

where every outer sum runs from $m = i$ to $L - 1$. □

C.3 The three root-flip tests

The weight in (C.8) is determined by the surviving set. To estimate the accompanying deletion probability, we hide the parts of clauses that are already satisfied at the fixed root values. They will test whether a root can be flipped.

The *type* of a baseline clause through index m records the roots it contains and the signs of its root literals. For $j \in R$, let $\mathcal{I}_{j,m}$ be the indices other than i , at most m , whose clause contains exactly root j and whose root literal is true at t ; put $s_{j,m} = |\mathcal{I}_{j,m}|$. Call a clause containing at least two roots a *collision*, and let v_m count such clauses at baseline indices through m . Finally, let $J_m \subseteq R$ contain those roots j for which no collision through m has exactly one root literal true under t , at root j . A collision can exclude at most one root, hence

$$|J_m| \geq 3 - v_m. \quad (\text{C.9})$$

Let $\mathcal{G}_m$ reveal all these prefix types and the entire signed nonroot part of every baseline clause through m , except at indices in $\bigcup_{j \in R} \mathcal{I}_{j,m}$. Thus $\mathcal{G}_m \subseteq \mathcal{F}_m$. The hidden clauses are already satisfied by their root literal at t , so their nonroot parts do not affect S_m . In particular $S_m, J_m, s_{j,m}, v_m$, and $\mathcal{W}_m$ are $\mathcal{G}_m$ -measurable. Conditional on $\mathcal{G}_m$, the hidden nonroot parts are independent uniform proper two-clauses on the u nonroots. Indeed, conditioning the independent clauses on their types separately preserves their product law, and revealing nonroot parts at other indices does not change it.

Put

$$d = \lceil \log n \rceil, \quad q_m = q_d(S_m).$$

Lemma C.6 (Root-flip tests under prefix conditioning). *For every $i \leq m < L$, on the $\mathcal{G}_m$ -measurable event $\{\max_{j \in R} s_{j,m} \leq d, |J_m| \geq 1\}$,*

$$\mathbb{P}(A_m \mid \mathcal{G}_m) \leq \mathbf{1}_{\{S_m \neq \emptyset\}} q_m^{|J_m|}. \quad (\text{C.10})$$

Proof. If A_m occurs, then for each $j \in J_m$ the pair clauses at indices in $\mathcal{I}_{j,m}$ must kill S_m . Otherwise choose an assignment in S_m satisfying those pairs, extend it with root values t , and flip only j . A clause with no true root literal before the flip has a satisfied nonroot part, which is unchanged. A clause with a true root literal at a root other than j remains satisfied. Any remaining clause has j as its unique true root before the flip. It cannot be a collision because $j \in J_m$, so it belongs to $\mathcal{I}_{j,m}$ and its pair is satisfied by the chosen assignment. The flipped assignment therefore satisfies B_m with root values different from t , contradicting (C.5).

The required killing events for different j use disjoint families of the conditionally independent hidden pairs. On a nonempty S_m , each family has at most d members and therefore kills with probability at most $q_d(S_m) = q_m$. A family of zero pairs has killing probability zero, so it causes no exception. Multiplication of these conditional bounds, together with the necessity of $S_m \neq \emptyset$, proves (C.10). $\square$

C.4 Truncating incidence counts and summing occupation

It remains to sum the right side of (C.8). The tests make deletion unlikely when q_m is small. When it is large, the lifetime estimate limits both the remaining deletion time and the number of indices at that level. We first isolate the rare prefixes for which too many tests or collisions are present.

Lemma C.7 (Prefix incidence counts). *Uniformly in $i \leq m < L$ and in the fixed value of C_i ,*

$$\mathbb{P}(\max_{j \in R} s_{j,m} > d) = O(n^{-4}), \quad (\text{C.11})$$

$$\mathbb{P}(v_m \geq 1) = O(n^{-1}), \quad \mathbb{P}(v_m \geq 2) = O(n^{-2}). \quad (\text{C.12})$$

Proof. A baseline clause contains a specified root with probability $3/n$, independently across indices, and $s_{j,m}$ is no greater than that root's incidence count. A union bound over roots and choices of d indices among at most $L = 10n$ trials gives

$$\mathbb{P}(\max_{j \in R} s_{j,m} > d) \leq 3 \binom{L}{d} (3/n)^d \leq 3(30e/d)^d = O(n^{-4}). \quad (\text{C.13})$$

The final bound holds for sufficiently large n , since $d = \lceil \log n \rceil$ and $\log d \rightarrow \infty$.

A trial contains any specified pair of roots with probability $6/[n(n-1)]$. The union over the three pairs bounds its collision probability by $18/[n(n-1)]$. A union bound over at most L independent trials gives $\mathbb{P}(v_m \geq 1) = O(n^{-1})$; a union bound over pairs of trials gives $\mathbb{P}(v_m \geq 2) \leq \binom{L}{2} O(n^{-4}) = O(n^{-2})$. $\square$

With $K = 64$, set

$$a_0 = \frac{4Kd}{u^2} = \frac{256d}{(n-3)^2}. \quad (\text{C.14})$$

For sufficiently large n , $a_0 < 1$. On $S_m \neq \emptyset$,

$$\mathcal{W}_m q_m^{3/2} \leq C d^{3/2}. \quad (\text{C.15})$$

For $q_m \geq a_0$, this follows by applying Proposition C.4 with $a = q_m$, and then using $\mathcal{W}_m \leq \tau(S_m)/p_0$. For $q_m < a_0$, including $q_m = 0$, use $\mathcal{W}_m \leq L$ and

$$L a_0^{3/2} = 10n \left(\frac{256d}{(n-3)^2} \right)^{3/2} = O(d^{3/2} n^{-2}) \leq C d^{3/2}. \quad (\text{C.16})$$

Lemma C.8 (Reduction to occupation). *For all sufficiently large n , uniformly in i and in the value of C_i ,*

$$\mathbb{E} D_i^2 \leq C(1 + d^{3/2}) + C d^{3/2} \sum_{m=i}^{L-1} \mathbb{E} [\mathbf{1}_{\{S_m \neq \emptyset\}} q_m^{3/2}]. \quad (\text{C.17})$$

Proof. In (C.8), the terms on $\{\max_j s_{j,m} > d\} \cup \{v_m \geq 2\}$ contribute at most

$$L \sum_{m=i}^{L-1} [\mathbb{P}(\max_j s_{j,m} > d) + \mathbb{P}(v_m \geq 2)] = O(L^2(n^{-4} + n^{-2})) = O(1).$$

For every remaining term, the restrictions, $\mathcal{W}_m$, and S_m are $\mathcal{G}_m$ -measurable. If $v_m = 1$, then $|J_m| \geq 2$. Conditioning on $\mathcal{G}_m$ in Lemma C.6 bounds its weighted probability by $\mathbf{1}_{\{S_m \neq \emptyset\}} \mathcal{W}_m q_m^2$. On a nonempty set, (C.15) and $q_m \leq 1$ bound this by $C d^{3/2}$. The one-collision contribution is therefore at most

$$C d^{3/2} \sum_{m=i}^{L-1} \mathbb{P}(v_m = 1) = O(d^{3/2}).$$

No independence between v_m and the surviving set is asserted or needed.

If $v_m = 0$, all three roots are available, and the same conditional bound has exponent three. By (C.15),

$$\mathbf{1}_{\{S_m \neq \emptyset\}} \mathcal{W}_m q_m^3 \leq C d^{3/2} \mathbf{1}_{\{S_m \neq \emptyset\}} q_m^{3/2}.$$

We may now discard the incidence restrictions. Summing the three cases and absorbing the factor three from (C.8) proves (C.17). $\square$

We next bound how long the decreasing sets S_m can remain nonempty at a given pair-killing level. The first visit to that level determines a set whose future lifetime controls all subsequent visits.

Lemma C.9 (Occupation above a level). *For all sufficiently large n and every $a_0 \leq a \leq 1$,*

$$\sum_{m=i}^{L-1} \mathbb{P}(S_m \neq \emptyset, q_m \geq a) \leq Cd^{3/2}a^{-3/2}. \quad (\text{C.18})$$

For $0 < a < a_0$, the same sum is at most L .

Proof. Let σ_a be the first index in $\{i, \dots, L-1\}$ for which $S_m \neq \emptyset$ and $q_m \geq a$, and set $\sigma_a = \infty$ when none exists. This is a stopping time for $\mathcal{F}_m$, because q_d is a deterministic function of its set argument. On $\{\sigma_a = m\}$, every counted index is at least m . Also $S_\ell = \emptyset$ for $\ell \geq m + U_m$: by then the future ordinary clauses have killed S_m , and any member of S_ℓ would be a member of S_m satisfying all those clauses. The number of counted indices is thus at most U_m , including the first index m .

For $a \geq a_0$, the event $\{\sigma_a = m\}$ is $\mathcal{F}_m$ -measurable, and on it the lifetime proposition applies. The fresh-future identity gives

$$\begin{aligned} \mathbb{E}[\mathbf{1}_{\{\sigma_a=m\}} U_m] &= \mathbb{E} \left[\mathbf{1}_{\{\sigma_a=m\}} \frac{\tau(S_m)}{p_0} \right] \\ &\leq Cd^{3/2}a^{-3/2} \mathbb{P}(\sigma_a = m). \end{aligned}$$

Summing over the disjoint first-visit events proves (C.18). This conditioning is legitimate because $m \geq i$ and all trials after m remain unexposed. At a smaller level there are at most L indices to count. $\square$

Proof of Theorem C.1. For $0 \leq q \leq 1$, $q^{3/2} = \int_0^1 \frac{3}{2} a^{1/2} \mathbf{1}_{\{q \geq a\}} da$. Applying this identity and Lemma C.9 gives

$$\begin{aligned} \sum_{m=i}^{L-1} \mathbb{E}[\mathbf{1}_{\{S_m \neq \emptyset\}} q_m^{3/2}] &= \int_0^1 \frac{3}{2} a^{1/2} \sum_{m=i}^{L-1} \mathbb{P}(S_m \neq \emptyset, q_m \geq a) da \\ &\leq La_0^{3/2} + Cd^{3/2} \int_{a_0}^1 \frac{da}{a} \leq Cd^{3/2} \log n. \end{aligned}$$

The final step uses (C.16) and $\log(1/a_0) \leq 2 \log n$, since $a_0 \geq n^{-2}$. The sum is finite and the integrands are nonnegative, so the interchange of sum, expectation, and integral is valid. Lemma C.8 now yields, uniformly in the deleted index and the fixed clause,

$$\mathbb{E}D_i^2 \leq C(1 + d^{3/2}) + Cd^3 \log n \leq C(\log n)^4. \quad (\text{C.19})$$

Average over C_i to remove the conditioning, and sum (C.19) in (C.4). This gives $\text{Var}(T_n) \leq Cn(\log n)^4$ for sufficiently large n . Enlarging C covers the remaining integers $n \geq 3$, since $0 \leq T_n \leq 10n$. Chebyshev's inequality proves the stated tail bound. $\square$

For every fixed $0 < \eta < 1/2$, the η and $1 - \eta$ quantiles of T_n therefore lie within $O_\eta(\sqrt{n} \log^2 n)$ of m_n . The probability outside any diverging multiple of this scale tends to zero. These statements use the finite-size mean m_n .

C.5 Cap changes and the center comparison

The prefix proof supplies a second concentration input for Theorem B.1. That theorem assumes a concentration bound and proves the center comparisons independently of how the bound is obtained. We apply it here using the variance bound of Theorem C.1, together with the cap identities already proved in Section 7. Put

$$W_n = \min\{H_n - 1, 10n\}, \quad V_n = \min\{H_n - 1, 16n\}, \quad e_n = \frac{\mathbb{E}T_n}{n}, \quad P_n(j) = \mathbb{P}(H_n > j).$$

Corollary C.10. *The variance and tail bounds of Theorem C.1 hold also for W_n and V_n , each centered at its own expectation. Moreover $e_n \rightarrow \alpha_3$, the strict-side threshold of the proper three-clause model: for every fixed $c \geq 0$ with $c \neq \alpha_3$,*

$$P_n(\lfloor cn \rfloor) \rightarrow \begin{cases} 1, & c < \alpha_3, \\ 0, & c > \alpha_3. \end{cases}$$

There is an absolute constant C such that for all sufficiently large $n = \min\{n_1, n_2\}$ with $\max\{n_1, n_2\} \leq 3n$, and for all sufficiently large n in the second inequality,

$$e_{n_1+n_2} \geq \min\{e_{n_1}, e_{n_2}\} - Cn^{-1/4}, \quad e_{n+1} \geq e_n - Cn^{-1/4}. \quad (\text{C.20})$$

Proof. Set $\rho = 2(7/8)^{10} < 1$. With $T = T_n$, $W = W_n$, and $V = V_n$, identities (7.1)–(7.2) show that the mean corrections from $m_n - 1$ are at most ρ^n and $(6n+1)\rho^n$, respectively. The corresponding absolute differences of standard deviations are at most $\rho^{n/2}$ and $(6n+1)\rho^{n/2}$; the sharper geometric bounds are those in Section 7, obtained from Proposition 5.1. These exponentially small corrections transfer the variance estimate to both last-satisfiable statistics. Chebyshev's inequality gives their tails about their own means.

For the center claim, choose a fixed $0 < c < e^{-2}$ and put $j = \lfloor cn \rfloor$. Lemma 6.1 gives $\mathbb{E}T_n \geq jP_n(j) = cn - o(n)$. The survival-sum bound (B.15) with $k = 3$ and density six gives $\mathbb{E}T_n \leq \mathbb{E}H_n \leq 6n + 8[2(7/8)^6]^n$. Thus $c_0 \leq e_n \leq 7 < 10$ eventually, for some absolute $c_0 > 0$. The relevant survival identity is $P_n(j) = \mathbb{P}(T_n > j)$ for integers $0 \leq j < 10n$; the restriction below the cap is essential, whereas the last-SAT identity $P_n(j) = \mathbb{P}(W_n \geq j)$ includes $j = 10n$. Theorem C.1 supplies the concentration input

$$\mathbb{P}(|T_n - m_n| \geq n^{3/4}) \leq \eta_n, \quad \eta_n = C \frac{(\log n)^4}{\sqrt{n}} \rightarrow 0. \quad (\text{C.21})$$

Apply Theorem B.1 with $k = 3$, $B = 10$, $a = c_0$, $A = 7$, and $\delta = 1/4$. It gives both comparisons in (C.20), in particular for the stated comparable sizes, and convergence with the proper strict-side limits. Uniqueness identifies the limit with α_3 of [16, Theorem 1.1]. The one-sided estimate (B.6) also gives, for every sufficiently large s and every $N \geq s^2$,

$$e_N \geq e_s - C's^{-1/4}.$$

□

For the comparable-size consequence, the coarser trial comparison gives a concrete error bound. With $N = n_1 + n_2$, $n = \min(n_1, n_2)$, $e_* = \min(e_{n_1}, e_{n_2})$, set $\varepsilon = n^{-1/4}$ and $M = \lfloor (e_* - 2\varepsilon)N \rfloor$, as

in the proof of Theorem B.1. The anchors give $(c_0/4)N \leq M \leq 7N$ for large n . Since $C_3 = 27$ in Lemma B.2 and $N \leq 4n$, (B.9) bounds the loss by

$$\frac{27M}{\varepsilon^3 n^3} \leq \frac{27 \cdot 7 \cdot 4n}{n^{9/4}} = 756n^{-5/4}. \quad (\text{C.22})$$

The restriction $\max(n_1, n_2) \leq 3n$ is used only for this numerical bound. For neighboring sizes, set $N = n + 1$ and $M = \lfloor m_n - 2n^{3/4} \rfloor$. The constant $D_3 = 729/8$ gives loss $(729/8)M/(\varepsilon^3 n^3) = O(n^{-5/4})$. In either case, the count-transfer index $j = \lfloor (1 - \varepsilon)M - N^{3/4} \rfloor$ is nonnegative for large n and lies strictly below $10N$, as required in (B.13).

The deviation bounds above are about each statistic's finite-size mean. The comparison inequalities are one-sided and do not give a two-sided rate for $e_n - \alpha_3$.

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